



NATIONAL AERONAUTICS AND SPACE ADMINISTRATION

REVISION A

APOLLO 13 FLIGHT PLAN

AS-508/CSM-109/LM-7

MARCH 27, 1970

FLIGHT PLANNING BRANCH
FLIGHT CREW SUPPORT DIVISION



MANNED SPACECRAFT CENTER
HOUSTON, TEXAS

UNITED STATES GOVERNMENT

Memorandum

TO : Distribution

DATE: MAR 23 1970

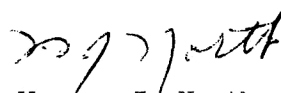
FROM : CF/Chief, Flight Crew Support Division

In reply refer to:
CF62-70M-089

SUBJECT: Revision A to the Apollo 13 Final Flight Plan

Enclosed is Revision A to the Apollo 13 Final Flight Plan. Revision A includes:

- a. Pen and ink changes to be made on indicated pages
- b. Pages to be inserted in place of original pages
- c. Attitude illustration pages to be inserted in section 3


Warren J. North

Enclosure

CF62:TRLindsey:avg 3-24-70



5010-108

Buy U.S. Savings Bonds Regularly on the Payroll Savings Plan

APOLLO FLIGHT DATA FILE

ORIGINAL 3/16/70

REVISION A 3/27/70

LIST OF EFFECTIVE PAGES

* INDICATES CURRENT CHANGES

PAGE NUMBER	ISSUE
i thru xx	Original
1-1 thru 1-25	Original
2-1 thru 2-4	Original
3i thru 3-54A	Original
*3-54B	Revision A
3-55 thru 3-56	Original
*3-56A	Revision A
3-57 thru 3-57A	Original
*3-57B	Revision A
3-58	Original
*3-58A	Revision A
3-59 thru 3-60	Original
*3-60A thru 3-60B	Revision A
3-61 thru 3-69	Original
*3-69A thru 3-69B	Revision A
3-70 thru 3-72	Original
*3-72A thru 3-72B	Revision A
3-73 thru 3-74A	Original
*3-74B	Revision A
3-75 thru 3-76	Original
*3-76A thru 3-76C	Revision A
3-77 thru 3-78A	Original
*3-78B	Revision A
3-79 thru 3-80B	Original
*3-80C	Revision A
3-81 thru 3-82B	Original
*3-82C	Revision A
3-83 thru 3-84A	Original
*3-84B	Revision A
3-85 thru 3-86	Original
*3-86AA	Revision A
3-86A thru 3-88	Original
*3-88A	Revision A
3-89 thru 3-95A	Original
*3-95B	Revision A
3-96 thru 3-97A	Original
*3-97B	Revision A
3-98 thru 3-99	Original
*3-99A	Revision A

3-100 thru 3-102A	Original
*3-102B	Revision A
3-103 thru 3-103A	Original
*3-104 thru 3-105	Revision A
3-106 thru 3-107	Original
*3-107A thru 3-107B	Revision A
3-108 thru 3-109	Original
*3-109A thru 3-109B	Revision A
3-110 thru 3-111	Original
*3-111A	Revision A
3-112 thru 3-113	Original
*3-113A	Revision A
3-114 thru 3-117	Original
*3-117A	Revision A
3-118 thru 3-119	Original
*3-119B	Revision A
3-120 thru 3-122	Original
*3-122A thru 3-122B	Revision A
3-123 thru 3-126	Original
*3-126AA	Revision A
3-126A	Original
*3-126B	Revision A
3-127 thru 3-128A	Original
*3-128B	Revision A
3-129 thru 3-130	Original
*3-130A	Revision A
3-131 thru 3-192	Original
4-1 thru 4-32	Original
5-1 thru 5-10	Original
6-1 thru 6-3	Original

Make the following pen and ink changes to the Final Apollo 13 Flight Plan dated March 16, 1970.

- Pg. 1-5 Paragraph 6, line 7 as reads "pitched up 20°" change to read "pitched up 90°".
- Pg. 1-14 Add on first blank line of TV schedule:
"Saturday, Apr. 11, 2:49 pm, 01:36, 05 min, EARTH SURFACE, CSM/S-IVB, MILA". Change Note at bottom of page to read
"***Approval received for Satellite time".
- Pg. 1-17 Line 6, change MCC-4 Refsmmat to "PTC".
- Pg. 1-22 Line 3, change 13-1 Altitude to -00.279; at bottom of page, change Mean Lunar Radius to "1738.09 KM".
- Pg. 1-24 In set TLC-1 (06:00), change 3rd Star (76) to read "125/EFH EPSILON SCORPII". In set TLC-2 (31:00), change 3rd Star (42) to read "212/EFH DELTA SAGITTARII". In set TLC-2 (31:00), change 4th Star (45) to read "44/ENH ENIF".
- Pg. 3-2 At 01:35, add "TV (MILA) 01:36 to 01:41".
- Pg. 3-3 At 02:45, MCC-H Column, add "DUMP DSE". At 02:55, fill in "OMNI C".
- Pg. 3-4 At 03:00, change DAP LOAD to read (11103, 11111).
- Pg. 3-4 At 03:10, fill in "HGA P -5, Y 297".
- Pg. 3-5 At 04:05, fill in "OMNI D".
At 04:25, add HGA P -10, Y 350.
At 04:35, delete "SECURE HGA . . .".
- Pg. 3-6 At 05:55, add "SECURE HGA, HGA TRACK-MAN, HGA Pitch -52, HGA Yaw 270".
- Pg. 3-7 At 06:00, fill in "OMNI B".
At 06:10, delete "OMNI ".
At 06:33, change 3rd Star (76) to read:
"3. STAR 125 EFH (R3 00120) EPSILON SCORPII
N88: (R1 -12736)(R2 -39323)(R3 -28133)".
- Pg. 3-9 Earth Field of VIEW, change GET to "10" hrs and F.O.V. to "10°".
- Pg. 3-10 Earth Field of VIEW, change GET to "8" hrs and F.O.V. to "11°".
- Pg. 3-25 At 31:00, add OMNI B, at 31:28 change 3rd Star (42, PEACOCK) to read:
"3. STAR 212 EFH (R3 00120) DELTA SAGITTARII
N88: (R1 +03605)(R2 -43220)(R3 -24881)".

At 31:33, change 4th STAR (45) to read:
"4. STAR 44 ENH (R3 00110) ENIF".

- Pg. 3-40 At 58:00, Fill in "Roll 285, HGA P 21, Y 270".
- Pg. 3-52 At 74:20, Change Moon View ATT to "R 359, P 111, Y 320,
HGA P -43, Y 283".
- Pg. 3-55 At 77:52, in notes column, change S-IVB LUNAR IMPACT NOTE to read "(GET 77:44) LAT 3.0°S, LONG 30.0°W".
- Pg. 3-60 At 82:00, in notes column add "NOTE" IF AUTO ACQ W/HGA unsuccessful, acquire in Manual Mode, Wide Beamwidth. If no comm use Omni A. At 82:04, center column, add "REPORT DOI ATTITUDE DEVIATIONS TO MSFN".
- Pg. 3-63 At 85:12, change "Tunnel Vent Valve - LM Press" to read "Tunnel Vent Valve - LM/CM ΔP".
- Pg. 3-73 At 99:43, CMP column, change "25 sec between marks" to read "10 sec between marks".
- Pg. 3-74 At 100:56, CMP column, change MNVR complete time to "(101:05)".
- Pg. 3-74A Change altitude of LANDMARK 13-1 as reads -000.18 to read "-000.28".
- Pg. 3-75 At 101:23, MCC-H column, after "UPLINK TO LM DESCENT TARGET" add "LPD BIAS (IF REQ'D)". At 101:30, LMP column, fill in "Steerable ANT P 132, Y -36".
- Pg. 3-75, 3-76 Add S/C darkness strip to LM MSFN column from 101:47:00 to 102:35:45. Terminator passage occurs at 101:42 and 102:41.
- Pg. 3-76 At 102:58, CMP column, delete "REACQUIRE MSFN V64" and add "OMNI B".
- Pg. 3-77 At 103:54, LMP column, delete "Biomed SW-RT"; at 103:57, MCC-H column, delete "Disable S-BD Relay"; at 103:00, MCC-H column, delete "DUMP DSE". At 103:22, MCC-H column add "DUMP DSE".
- Pg. 3-78 At 104:55, MCC-H column, add "Update to LM, P 22 ACQ TIME 28° ELEV". At 104:56, CMP column, add "ORBITAL SCIENCE" and at 104:59 add "PHOTO TGT 14".
- Pg. 3-78A 13-1 P22 LDMK TRACKING PAD, delete DAC shutter speed "(1/60)". Change 13-1 altitude as reads "-000.18" to read "-000.28".
- Pg. 3-79 At 105:00, CMP column, delete "ORBITAL SCIENCE, PHOTO TGT 14". At 105:23, MCC-H column, delete "P22 ACQ time 28° EL". At 105:59, CMP column, add "HGA P -6, Y 181; in MCC-H column add "DUMP DSE".
- Pg. 3-80 LM columns, delete line now showing "EQUIPMENT PREP" beginning at 106:30, draw in new line at 106:15.

- Pg. 3-81 At 107:19, CMP column, change TOPO Attitude to "R 358, P 123, Y 359".
- Pg. 3-82 At 108:02, CMP column, after "P 018" add "OMNI D".
- Pg. 3-85 At 111:36, CMP column, after "DAC OFF . . ." add "LOAD DAP (10102) (11111)".
- Pg. 3-87 At 113:52, CMP column, change MNVR COMPLETE TIME TO "(113:54)".
At 113:56, CMP column, delete "OMNI B" add HGA P -44, Y 225.
- Pg. 3-88 LM columns, delete line now showing "PLSS RECHARGE" beginning at 114:09. Draw in new line a 114:00. Delete "EAT PERIOD" in CDR and LMP columns at 114:04. At 114:54, change "SC CONT-CMC/FREE . . ." to read "SC CONT-CMC/AUTO . . .". At 115:00, delete "SC CONT CMC/AUTO".
- Pg. 3-94 At 124:24, LM column, change "LGC PUMP CB-CLOSE" to "LCG PUMP CB-CLOSE".
- Pg. 3-97 At 127:10, MCC-H column, add "MSFN record CMP voice on DSE". At 127:12, CMP column, add "OMNI A".
- Pg. 3-98 At 128:18, across LM column, add "PERFORM THERMAL DEGRADATION TEST".
At 128:35, CMP column, change "OMNI D" to "OMNI C".
- Pg. 3-99 At 129:22, CMP column, delete "V64 ACQ MSFN".
- Pg. 3-100 At 130:33, CMP column, change "OMNI C" to "OMNI D". At 130:41, CMP column, add "OMNI A". At 130:46, CMP column, add "V64 ACQ MSFN".
- Pg. 3-101 At 131:14, CMP column, fill in R 012, P 116.8, Y 008. At 131:25, CMP column, change "OMNI C" to "OMNI A".
- Pg. 3-103A 13-1 P22 LDMK Tracking Pad, delete DAC shutter speed "(1/60)". Change 13-1 altitude as reads "-000.18" to read "-000.28".
- Pg. 3-104 At 134:30, CMP column, change "OMNI C" to "OMNI D".
- Pg. 3-106 CMP column, at 136:30, change "STOW JETTISON ITEMS" to "PACK JETTISON ITEMS".
- Pg. 3-114 At 144:21, change "OMNI D" to "OMNI C". At 144:57, add "Presleep checklist". At 144:57, add "CABIN FANS (2)-OFF; remove and stow (on A-13) Cabin Fan Lunar Dust Filter".
- Pg. 3-115 At 145:18, delete "(10111, 11111), V21N01, 3255E, 1616E".
Add V79 $\begin{pmatrix} -0.0000 \\ +010.00 \\ +11111 \end{pmatrix}$
- Pg. 3-119 At 153:08, delete "MNVR TO P52 ATT (153:15), R 098, P 274, Y 315".
At 153:14, change HGA to "P -5, Y 250". At 153:43, delete "OMNI A".
- Pg. 3-120 At 154:55, change P52 ATT to "R 180, P 0, Y 320, OMNI A".

- Pg. 3-121 At 155:46, as reads "MNVR TO PHOTO ATT BY", delete "BY".
- Pg. 3-122 At 156:19, delete "V64 Acquire MSFN". At 156:48, add "V64 Acquire MSFN". At 156:53, change "VISUAL TGT 16, N . . ." to read "VISUAL TGT 15, N . . .".
- Pg. 3-124 At 158:32, add "V64 Acquire MSFN". At 158:47, delete "V64 Acquire MSFN". At 158:52, change f stop for TGT 63 from "f4" to "f5.6".
- Pg. 3-126 At 160:16, add "V64 Acquire MSFN". At 160:21, MCC-H column, after "Record PCM . . . ON DSE" add "(IF NO HGA)". At 160:35, delete "V64 Acquire MSFN". At 160:50, change f stop for TGT 66 from "f4" to "f5.6".
- Pg. 3-131 At 165:50, change TV Attitude Yaw to "Y 352".
- Pg. 3-136 At 170:03, add "OMNI D".
- Pg. 3-144 At 183:00, change Optics Cal Attitude to "R 075, P 063, Y 0" and add HGA P -79, Y 290. At 183:06, change Optics Cal Star to "STAR 4 4".
- Pg. 3-145 At 184:00, after "START PTC" add "S-BD ANT-OMNI B ON MCC CUE, SECURE HGA".
- Pg. 3-148 At 187:00, add "OMNI D".
- Pg. 3-152 At 191:00, add "OMNI B".
- Pg. 3-156 At 195:00, add "OMNI D".
- Pg. 3-164 At 208:00, add "OMNI B".
- Pg. 3-168 At 212:00, add "OMNI B".
- Pg. 3-173 At 217:00, add "OMNI B".
- Pg. 3-174 At 218:22, delete "R ____, P ____, Y ____".
- Pg. 3-176 At 220:00, add "OMNI B". At 220:54, fill in R 340, P 270, Y 0, HGA P 20, Y 270".
- Pg. 3-178 At 222:03, after "START PTC", add "S-BD ANT-OMNI B ON MCC CUE, SECURE HGA".
- Pg. 3-184 At 232:00, add "OMNI B".
- Pg. 3-187 At 235:50, fill in "R 075, P 063, Y 0, OMNI B OPTICS CAL STAR 4 4".
- Pg. 3-188 At 236:00, fill in "R 090, P 171, Y 345".
- Pg. 3-190 At 238:52, fill in "R 0, P 156, Y 0, OMNI B, VHF ANT RT".
- Pg. 3-192 At 240:35, fill in "R 0, P 156, Y 0".

CSM GNC GO CRITERIA

SPS

- 0 FU/OX TANK (W/O LEAK) -FU/OX
- 0 GN₂ TANK (W/O LEAK) - 1 OF 2 (CANT CONFIRM)
- 0 BALL VALVE BANK - BOTH
- 0 FEEDLINE TEMP >40° F
- 0 FU/OX ΔP<20 PSI
- 0 PC>70 PSI
- 0 ULLAGE CAPABILITY - 1 OF 2
- 0 HE TANK (W/O LEAK)

SM RCS

- 0 HE TANK (W/O LEAK) - ALL
- 0 NO LEAK BELOW ISO VLV - ALL
- 0 PKG TEMP > 55° - ALL
- 0 THRUSTERS - 3 OF 4 P & Y, 6 OF 8 R

CM RCS

- 0 HE TANK (W/O LEAK) - BOTH
- 0 MANIFOLD (W/O LEAK) - BOTH
- 0 NOT ARMED

SYSTEMS MANAGEMENT

- 0 START BURN ON BANK B
- 0 OPEN BANK A TIG + 2-5 SECONDS

LOI INHIBIT CRITERIA

- 0 FULL CRITICAL SYSTEMS REDUNDANCY
- 0 ADEQUATE CONSUMABLES FOR MIN LO OPS, CAPABILITY TO SUSTAIN TANK LOSS AND RETURN TO EARTH WITH AVG POWER LEVEL OF 40 AMPS
- 0 SPS PRPLNT RESERVE FOR TEI AND TEC MCC'S
- 0 RCS PRPLNT RESERVE FOR TEI AND TEC MCC CONTROL, PTC, AND MIN LUNAR ORBIT OPERATIONS
- 0 DPS LOI IF REQUIRED TO ACCOMPLISH A LUNAR ORBIT OPERATION

CSM GNC GO CRITERIA
GNCS/SCS

- 0 3 - AXIS AUTO ATTITUDE CONTROL
- 0 3 - AXIS RATE DAMPING
- 0 3 - AXIS DIRECT RCS
- 0 BMAGS P, Y -- 1 OF 2
- 0 BMAGS R -- 1 OF 2
- 0 FDAI -- 1 OF 2
- 0 CMC, ISS, OSS, OPTICS DAC
- 0 TVC SERVO LOOP -- BOTH
- 0 DSKY -- 1 OF 2

FLIGHT PLAN

LOI
BURN TABLE

P OR Y RATES	ATT DEVIATION	SHUTDOWN TIME	RESIDUALS
10°/SEC TAKEOVER	+10° TAKEOVER	BT + 10 SEC	DO NOT TRIM

TABLE 3-6
LOI ABORT TABLE

MODE I	MODE II	MODE III
(DPS ONLY)	(DPS ONLY)	(DPS ONLY)
0-1 MIN 45 SEC	1 MIN 45 SEC TO 2 MIN 50 SEC	3 MIN 50 SEC 5 MIN 57 SEC (CUTOFF)
ΔV_m 0-750 (TIGHT)	ΔV_m 750-1245 (LOOSE)	ΔV_m 1700-2815 (TIGHT)
LOI + 2 HR.	DPS_1 @ LOI + 2 HR	DPS @ LOI + 1 REV
MCC-H TARGET OR CREW CHART (NO COMM)	DPS_2 @ LOI + 1 REV MCC-H TARGET	DPS @ LOI + 1 REV MCC-H TARGET

TABLE 3-7

3-54B

REVISION A

TIGHT SPS LIMITS

FUEL - OXID DELTA P GREATER THAN 20 PSI
CONFIRMED BY LOW PC
PROP TANK PRESS LESS THAN 160 PSI CON-
FIRMED BY LOW PC
PC LESS THAN 80 PSI OR DECAYS 10 PSI
DURING THE BURN

ANY BALL VALVE FAILS TO OPERATE, OR
CLOSES PREMATURELY, SHUT DOWN THE
GOOD BANK AND (A) IF THRUST CEASES-
ABORT (B) IF THRUST CONTINUES-
REENABLE BOTH BANKS AND CONTINUE

NOTE: IF THE FIRST BANK SELECTED FAILS TO
OPERATE UNDER G & N CONTROL, ATTEMPT
TO START THAT BANK UNDER SCS CONTROL.
IF THE BANK STARTS UNDER SCS CONTROL,
CONTINUE THE BURN AND EVALUATE G & N
STEERING. IF THE BANK FAILS TO START
UNDER SCS CONTROL, INHIBIT LOI.

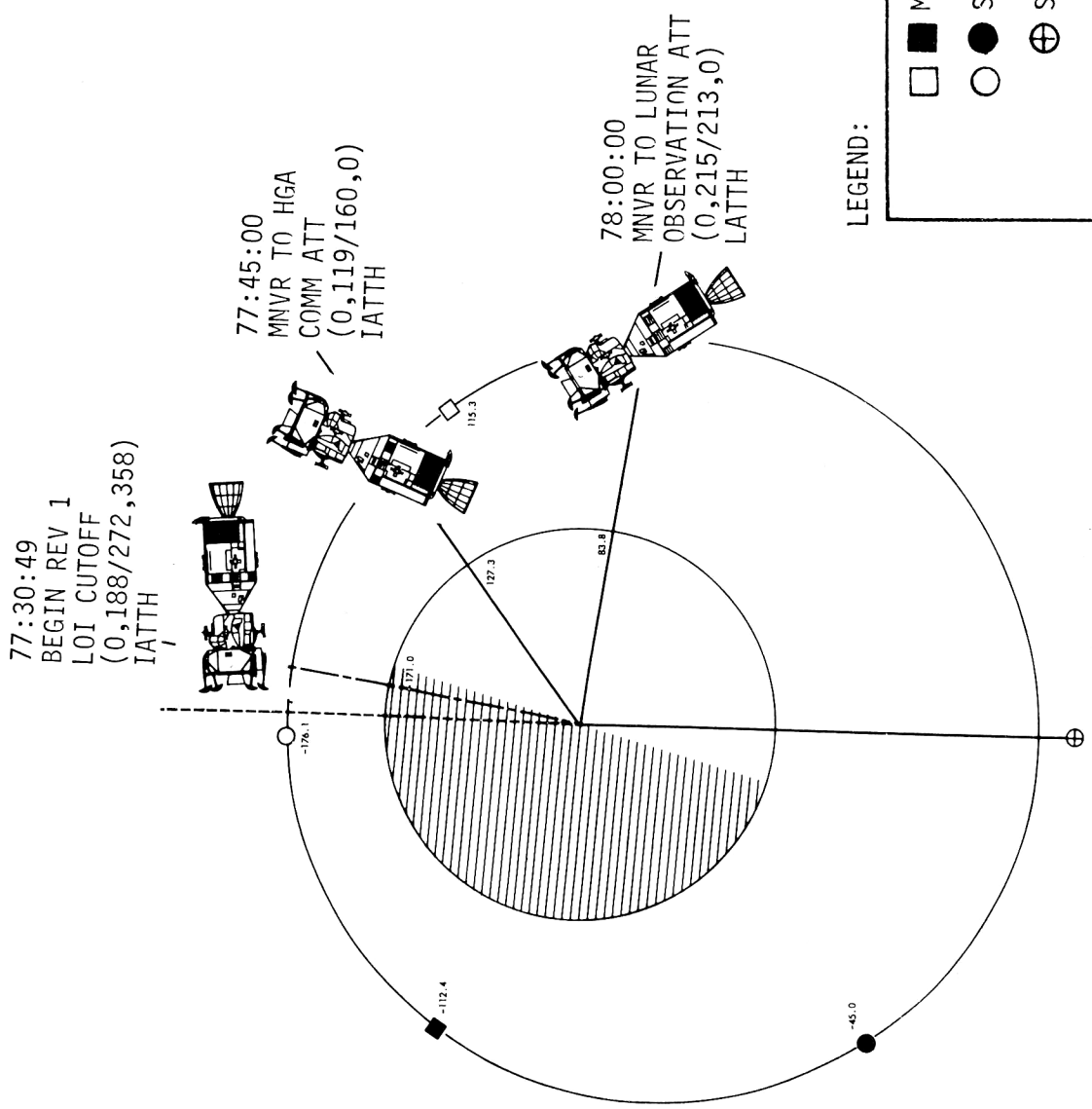
LOOSE SPS LIMITS

PC LESS THAN 70 PSI CONFIRMED BY OTHER
CUES

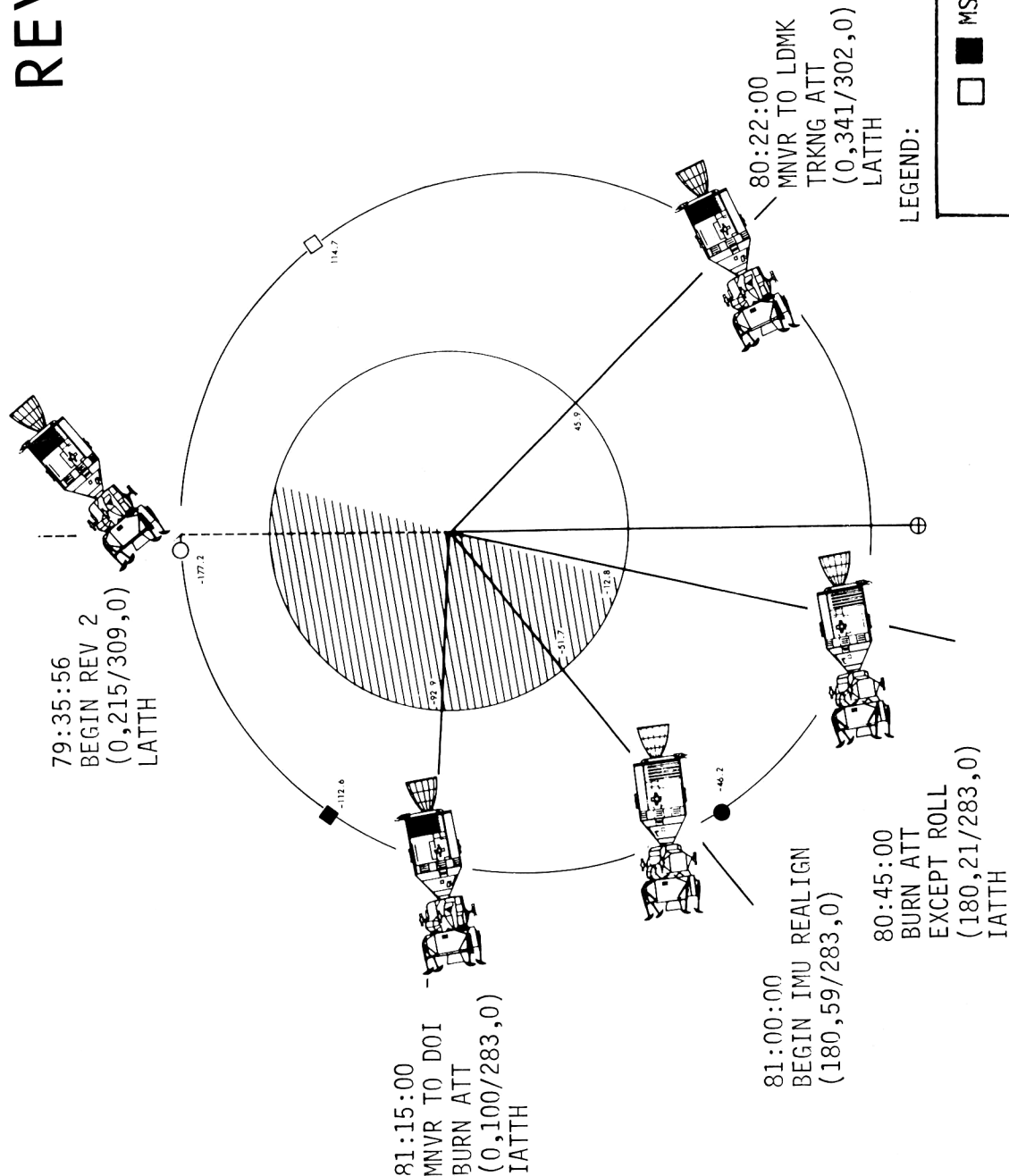
PROP TANK PRESS LESS THAN 115 PSI CONFIRM-
ED BY LOW PC

PHYSIOLOGICAL INDICATIONS OF ERRATIC
ENGINE PERFORMANCE (VIBRATION, POPPING,
ETC.)

REV 1

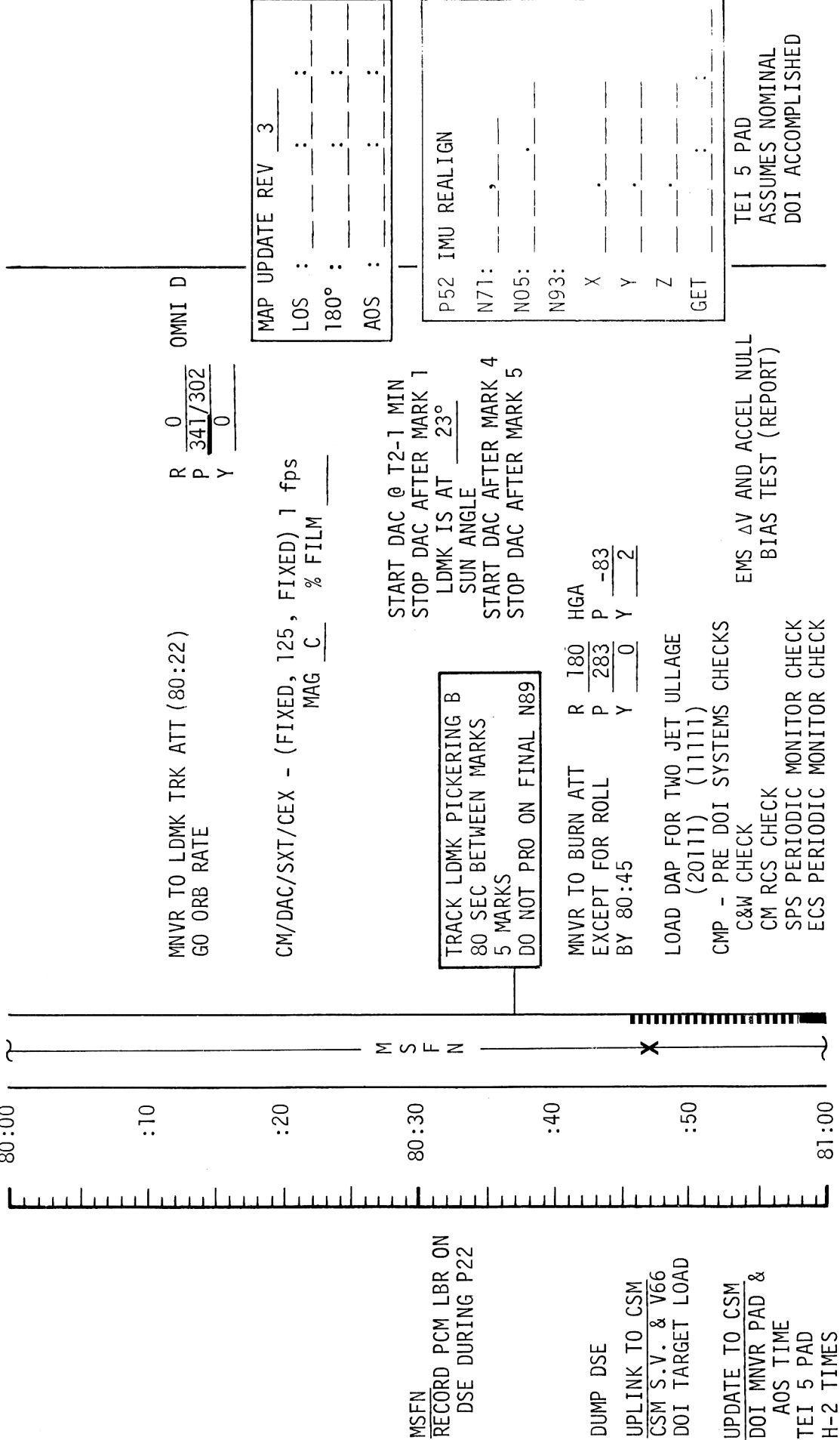


REV 2



3-57B

REVISION A



MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 13	FINAL (APRIL)	MARCH 16, 1970	80:00 - 81:00	4/2	3-58

FLIGHT PLAN

DOI

BURN TABLE

P OR Y RATES	ATT DEVIATIONS	SHUTDOWN TIME	RESIDUALS
10°/SEC TAKEOVER	+ 10° TAKEOVER	BT + 1 SEC	TRIM X TO WITHIN 1 FPS * DO NOT TRIM Y & Z

*IF OVERBURN IS >2.2 FPS PITCH 180 AND TRIM RCS IF <10, SPS IF >10.

AT AOS AFTER DOI, THE REQUIREMENT FOR THE BAILOUT MANEUVER WILL BE DETERMINED BY EVALUATION OF THE THREE TRAJECTORY MONITORING SOURCES. THESE SOURCES--PGNS, EMS, AND MSFN--WILL BE EXAMINED WITH THE FOLLOWING CRITERIA:

1. IF MSFN RADAR DATA IS VALID AND REASONABLE, A STAY VOTE FROM MSFN IS REQUIRED TO REMAIN IN THE LOW ORBIT.

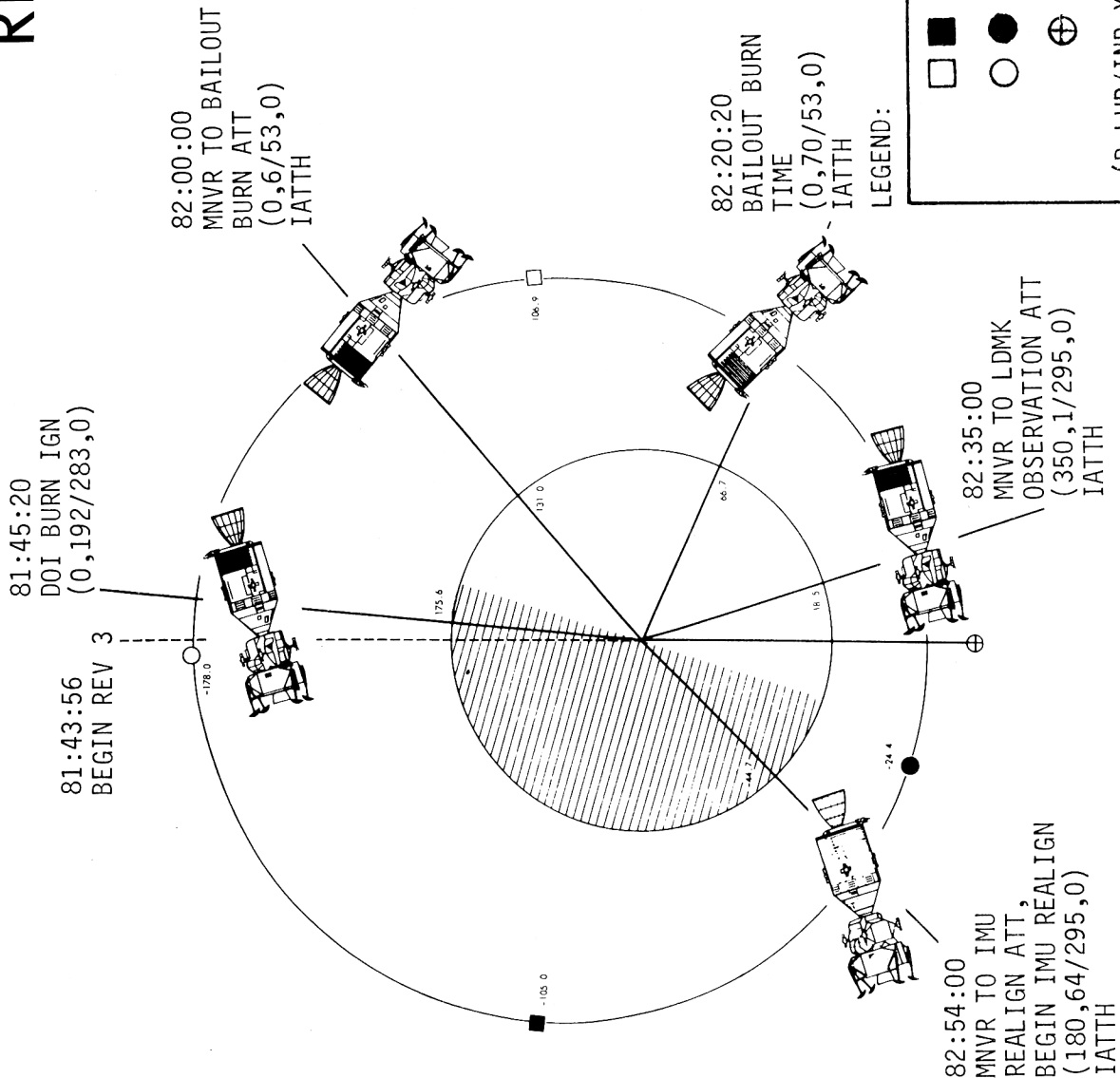
2. IF MSFN RADAR DATA IS INVALID OR UNAVAILABLE, THE FOLLOWING CRITERIA APPLIES:

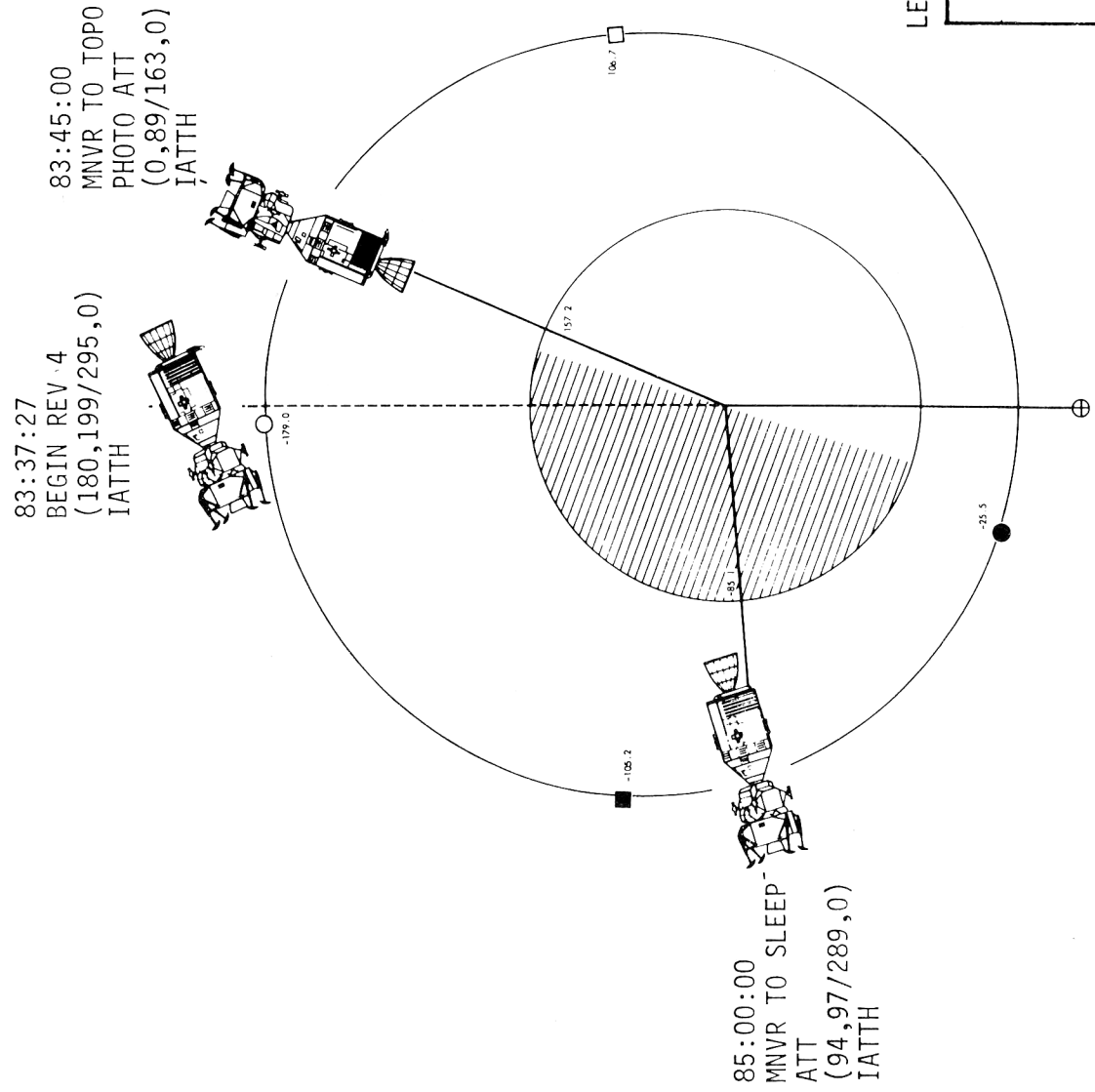
- A. 2 OUT OF 2 STAY VOTES REQUIRED TO REMAIN IN THE LOW ORBIT.
- B. 1 OF 1 STAY VOTE REQUIRED TO REMAIN IN THE LOW ORBIT.

(1) THE EMS VOTE IS NO STAY IF THE EMS INDICATES A 12FPS OVERSPEED AFTER TRIMMING THE PGNS.

(2) THE MSFN VOTE IS NO STAY IF THE INCOMING RADAR DATA INDICATES CLOSEST APPROACH ALTITUDE OF EQUAL TO OR LESS THAN 1.0 N MI ABOVE THE LUNAR TERRAIN. THIS ALTITUDE CORRESPONDS TO A PERICYNTHION ALTITUDE OF 3.6 N MI AND DOPPLER RESIDUALS AT AOS OF -32 CYCLES PER SECOND.

REV 3





LEGEND:

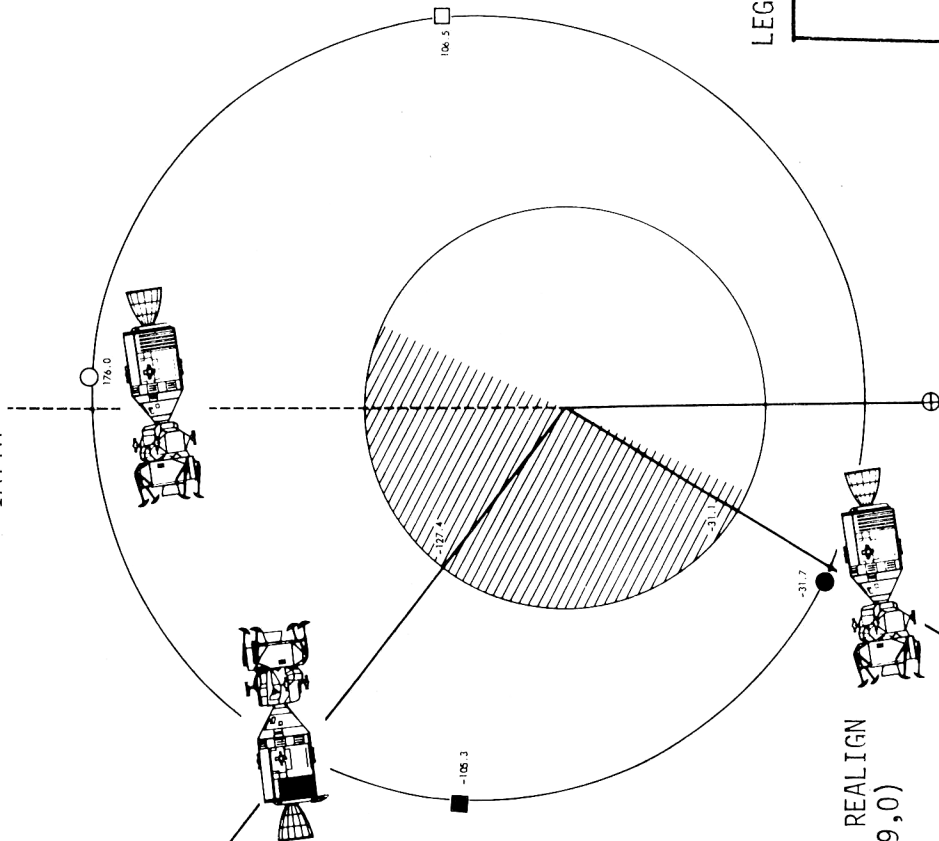
■	MSFN AOS, LOS
●	S/C SUNRISE, SUNSET
⊕	SUBEARTH POINT
(R,LHP/INP,Y)	
IATTH - INERTIAL ATTITUDE HOLD	
LATTH - LOCAL ATTITUDE HOLD	

REV 10

94:58:56
BEGIN REV 10
(94,187/289,0)
IATTH

96:35:00
MNVR TO AGS
CALIBRATION ATT
(8,316/113,23)
IATTH

96:05:00
BEGIN IMU REALIGN
(94,37/289,0)

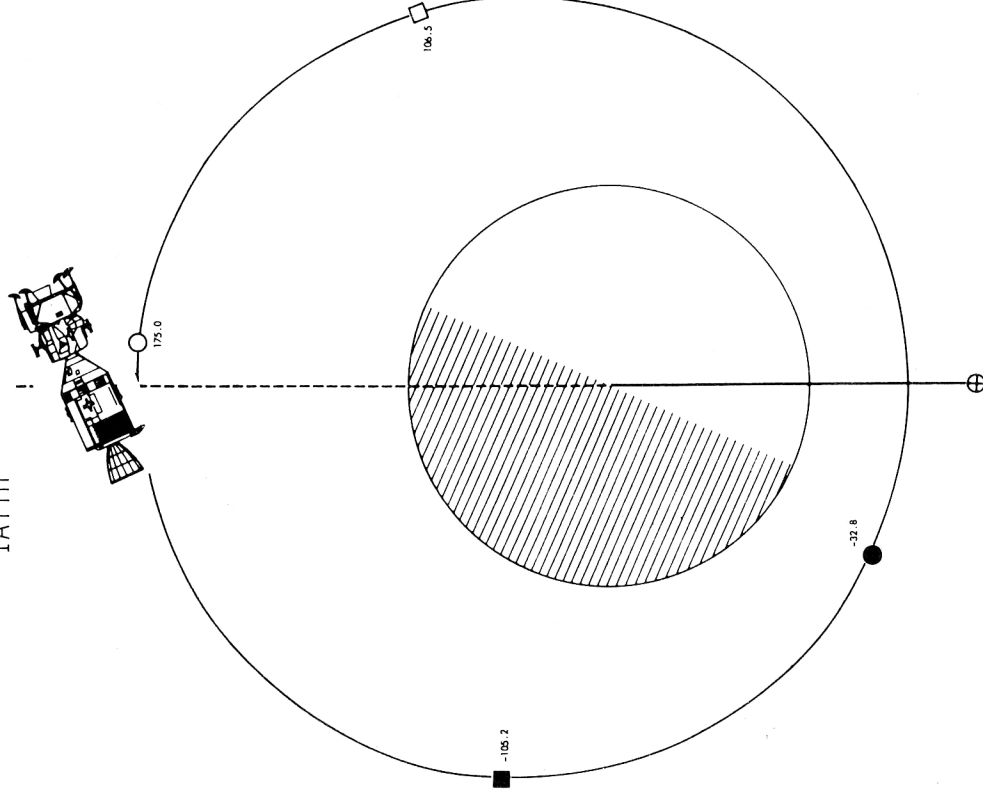


LEGEND:

- MSFN AOS, LOS
- S/C SUNRISE, SUNSET
- SUBEARTH POINT
- $(R, LHP/INP, Y)^*$
- IATTH - INERTIAL ATTITUDE HOLD
- LATTH - LOCAL ATTITUDE HOLD

96:52:30
BEGIN REV 11
(8,9/113,23)
IATTH

REV 11



LEGEND:

□	MSFN AOS, LOS
○	S/C SUNRISE, SUNSET
⊕	SUBEARTH POINT
(R,LHP/INP,Y)	
IATTH - INERTIAL ATTITUDE HOLD	
LATTH - LOCAL ATTITUDE HOLD	

3-69B

REVISION A

FLIGHT PLAN

CSM

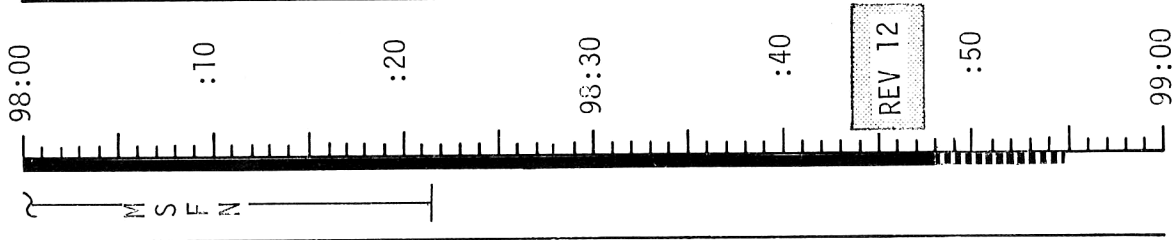
CMP

INHIBIT ROLL COMMANDS
UNTIL LM/CM ΔP>3.5 PSID
REMOVE & STOW
CSM/LM UMBILICAL
INSTALL DROGUE & PROBE
PRELOAD PROBE
COCK LATCHES (12)
INSTALL HATCH
VENT TUNNEL
HATCH INTEGRITY
CHECK
CMC - FREE/AUTO
VERIFY DSE MOTION AT LOS
RECORD LM PCM DATA
DOFF HELMET & GLOVES
LiOH CANISTER CHANGE
10 INTO B, STOW 8 IN B6
RR TRANSPONDER ACT
& SELF TEST

LOAD DAP
R=21101, R2=11111

V06N20E
INHIBIT B3
THRUSTER FOR
RR ACT & SELF TEST
CMC FREE FOR AGS CAL
P30/P41 TO MANEUVER
TO UNDOCKING ATT
(99:10)
R 0, P 102, Y 0

1513 CST



LM

CDR

DOCKED IMU FINE ALIGN

VERIFY DROGUE
& PROBE
INSTALLATION

CLOSE AND SECURE
HATCH

DON HELMET & GLOVES

LMP

COPY UPDATES
AGS ACTIVATION & SELF
TEST
AGS TIME INITIALIZATION
LOAD AGS PAD

STEERABLE ANT
P 131, Y 41
VHF B XMTR - DATA
DON HELMET & GLOVES

UPDATE TO LM
GYRO TORQUING X's
AGS ABORT CONSTANTS
DAP DATA
UPLINK TO LM
LS REFMMAT
LM S.V. & V66
LGC/CMC CLOCK SYNC
LGC ABORT CONSTANTS
E MEMORY UPDATE
(IF REQ'D)
UPDATE TO LM
AGS K FACTOR

MCC-H

MAP	UPDATE	REV	12
LOS	:	:	:
180°	:	:	:
AOS	:	:	:

ARS/PGA PRESSURE INTEGRITY CHECK

CABIN REGULATOR CHECK

DOFF HELMETS & GLOVES (CREW OPTION)

DRIFT CHECK
V06 N20E ON MARK

RATE GYRO
TEST

V47-AGS UPDATE & ALIGN

RR ACT & SELF TEST

DOCKED AGS
CALIBRATION

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 13	FINAL (APRIL)	MARCH 16, 1970	98:00 - 99:00	5/11-12	3-72

FLIGHT PLANNING BRANCH

CSM LOW ALTITUDE LANDMARK TRACKING PROFILE

Diagram illustrating the CSM Low Altitude Landmark Tracking Profile. The profile shows a circular path with six landmarks (MARK NO 1 to MARK NO 6) and their respective time-to-arrival (TCA) and time intervals (ΔT).

Key parameters and landmarks:

- MARK NO 1 (TCA-20 SEC)
- MARK NO 2 (TCA-10 SEC)
- MARK NO 3 (TCA)
- MARK NO 4 (TCA+10 SEC)
- MARK NO 5 (TCA+20 SEC)
- MARK NO 6 (TCA+20 SEC)

Time intervals (ΔT):

- $\Delta T_1 = 9^\circ$
- $\Delta T_2 = 17^\circ$
- $\Delta T_3 = 17^\circ$
- $\Delta T_4 = 47^\circ$
- $\Delta T_5 = 47^\circ$
- $\Delta T_6 = 17^\circ$

Other parameters:

- PITCH RATE: $2^\circ/\text{SEC}$ (TCA-50 SEC)
- HORIZON (TCA-128 SEC)
- RADIUS OF MOON

Table 1: P22 LDMK TRACKING (1/60)

T ₁	T ₂	R	N or S	NM	SA	TA	N89
13-1							

Table 2: Tracking Data

LAT	LONG/2	ALT
-04.043	-07.799	-000.28

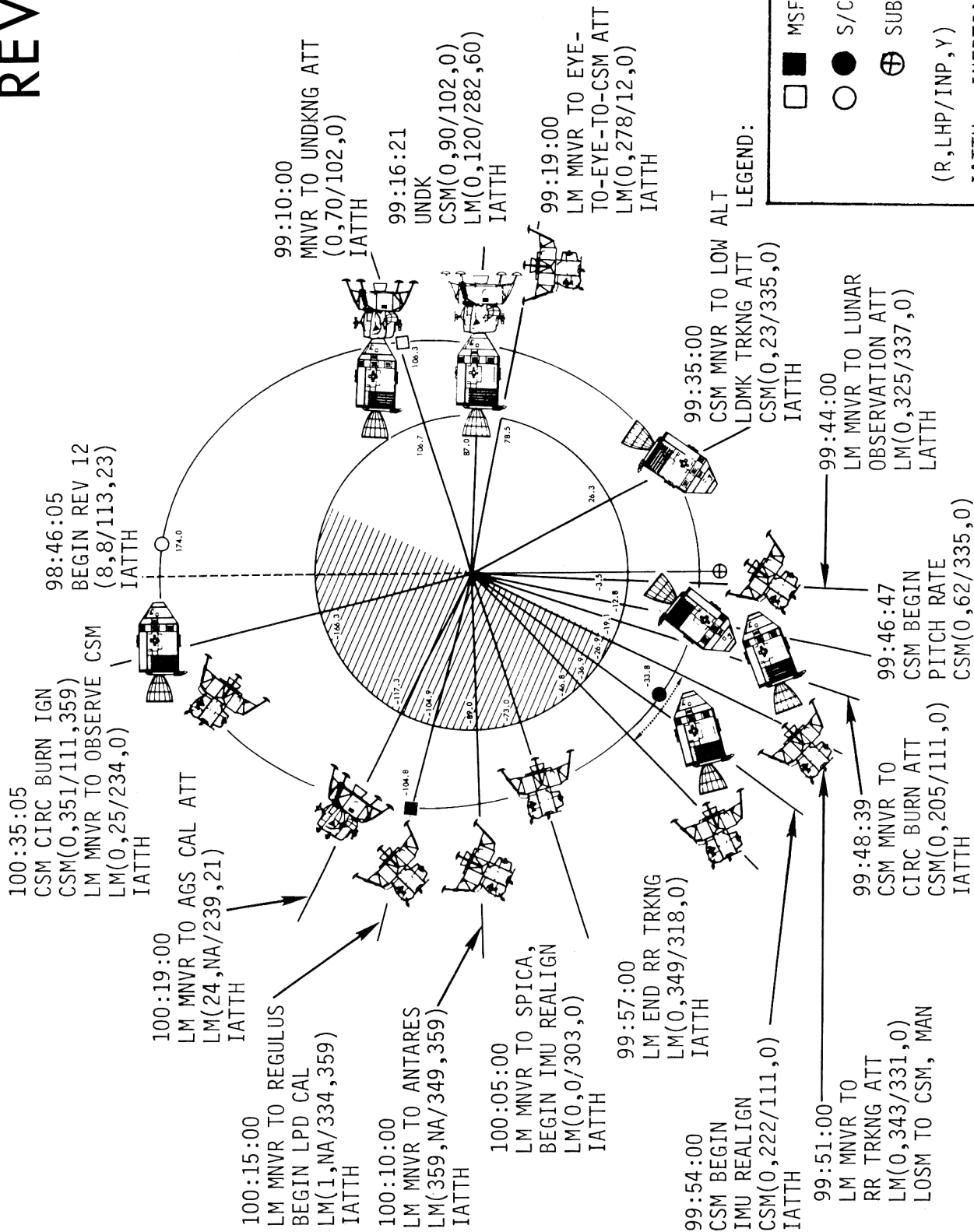
Table 3: Time Intervals

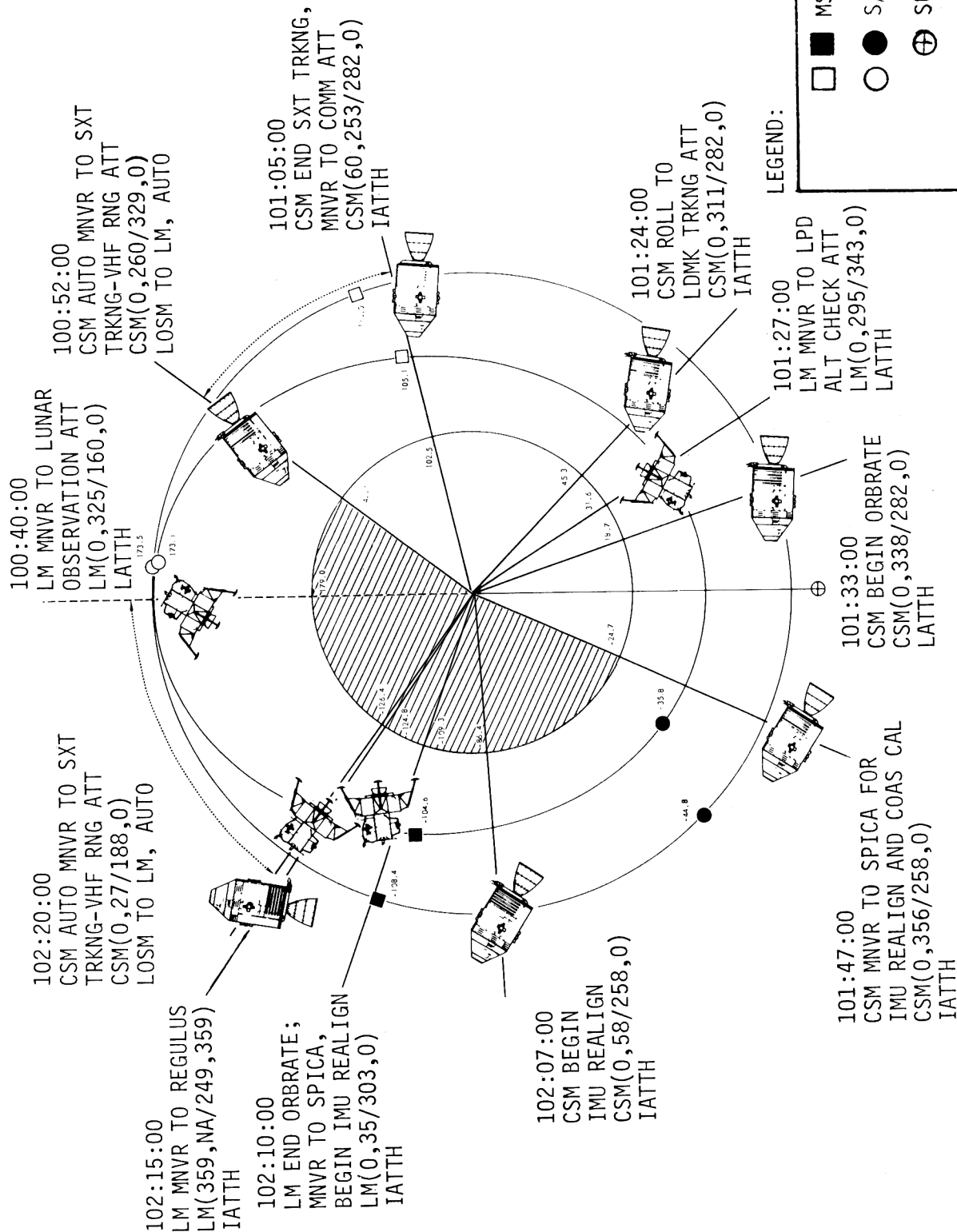
ΔT_1	ΔT_2	ΔT_3	ΔT_4	ΔT_5	ΔT_6
78 SEC	30 SEC	10 SEC	10 SEC	10 SEC	10 SEC

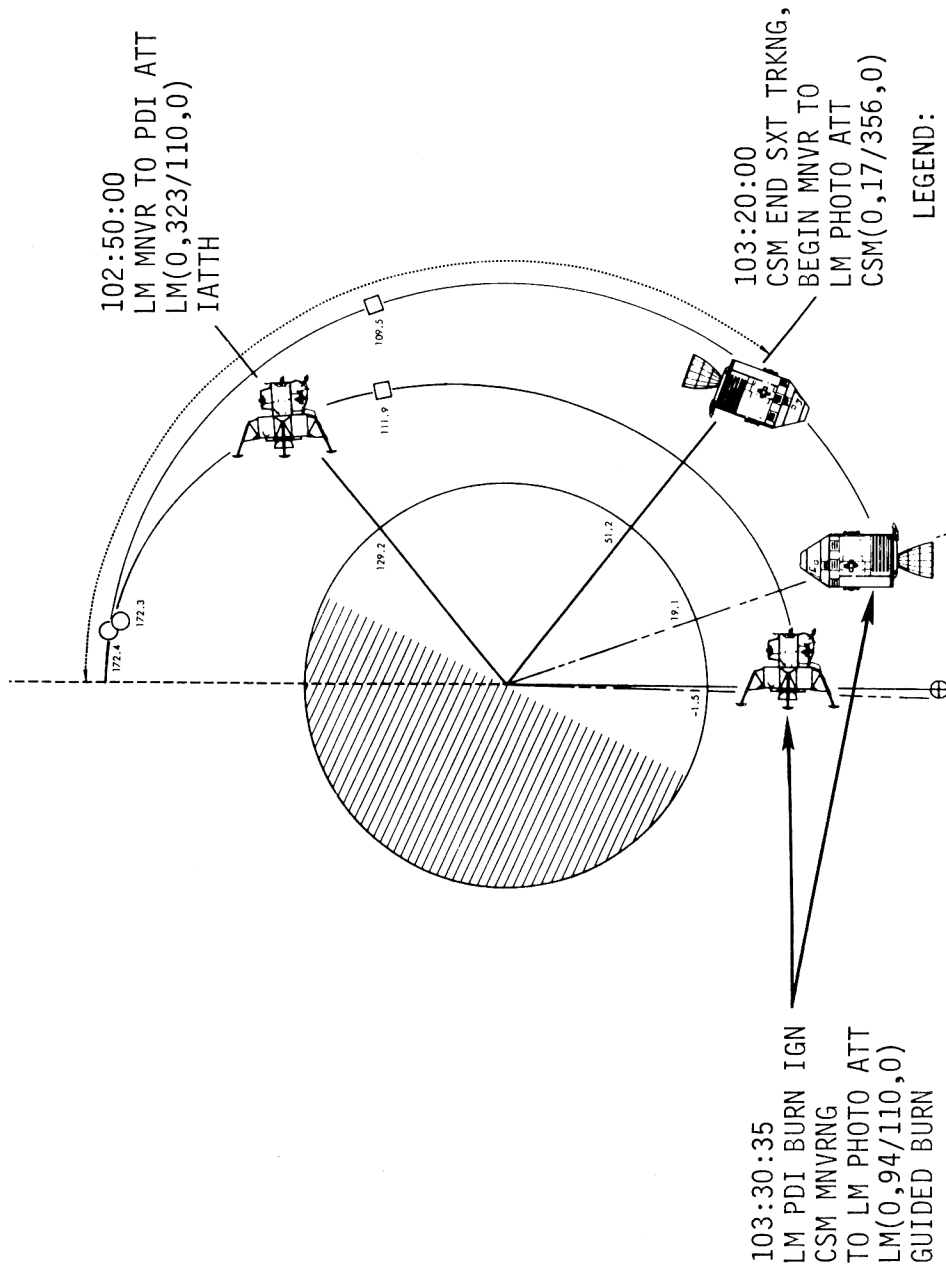
P22 LDMK TRACKING		(1/60)				
T ₁	_____	•	•	•	•	•
T ₂	_____	•	•	•	•	•
R	_____	°P	_____	SA	_____	°Y
N or S	NM	_____	_____	_____	_____	_____
LAT	-04.043	_____	_____	_____	_____	_____
LONG/2	-07.799	_____	_____	_____	_____	_____
ALT	-000.28	_____	_____	_____	_____	_____
		13-2	13-3	13-4	13-5	
LAT		-03.606	-03.189	-03.707	-03.226	
LONG/2		-07.658	-07.739	-07.006	-07.057	
ALT		+000.00	-000.76	-000.73	-000.85	

CENTER OF MOON
FIGURE 3-3
3-72A

REVISION A







LEGEND:

- MSFN AOS, LOS
 - S/C SUNRISE, SUNSET
 - ⊕ SUBEARTH POINT
- (R,LHP/INP,Y)
- IATTH - INERTIAL ATTITUDE HOLD
LATTH - LOCAL ATTITUDE HOLD

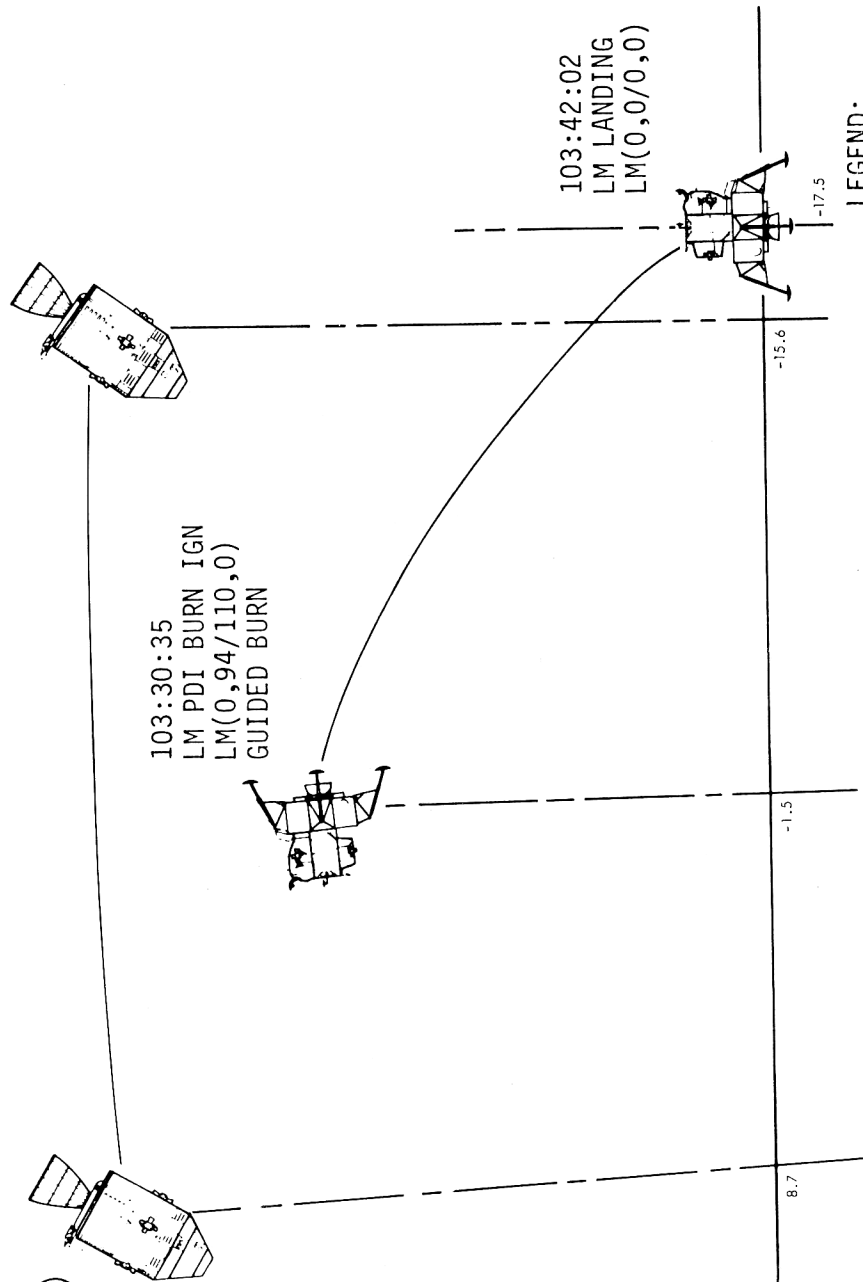
3-76A

103:42:02
LM LANDING
CSM(0,241/153,0)
IATTH

103:34:00
CSM MNVR TO
LM PHOTO ATT
CSM(0,217/153,0)
IATTH

103:30:35
LM PDI BURN IGN
LM(0,94/110,0)
GUIDED BURN

103:42:02
LM LANDING
LM(0,0/0,0)



LEGEND:

- MSFN AOS, LOS
 - S/C SUNRISE, SUNSET
 - ⊕ SUBEARTH POINT
- (R,LHP/INP,Y)
- IATTH - INERTIAL ATTITUDE HOLD
LATTH - LOCAL ATTITUDE HOLD

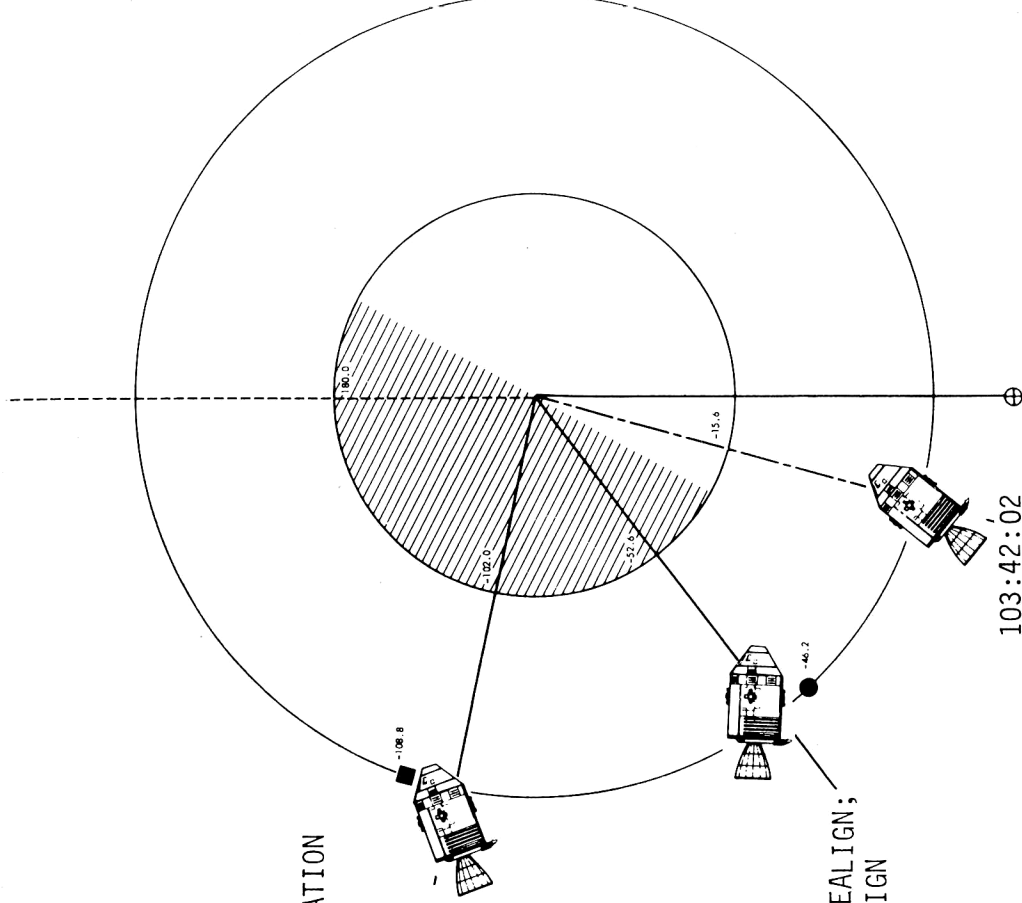
3-76B

REVISION A

104:10:00
MNVR TO SC CONTAMINATION
PHOTO ATT
(320,307/133,348)
IATTH

103:54:00
MNVR FOR IMU REALIGN;
BEGIN IMU REALIGN
(0,225/100,0)
IATTH

103:42:02
LM LANDING
(0,241/153,0)
IATTH



LEGEND:

□	MSFN AOS, LOS
○	S/C SUNRISE, SUNSET
⊕	SUBEARTH POINT
(R,LHP/INP,Y)	
IATTH -	INERTIAL ATTITUDE HOLD
LATTH -	LOCAL ATTITUDE HOLD

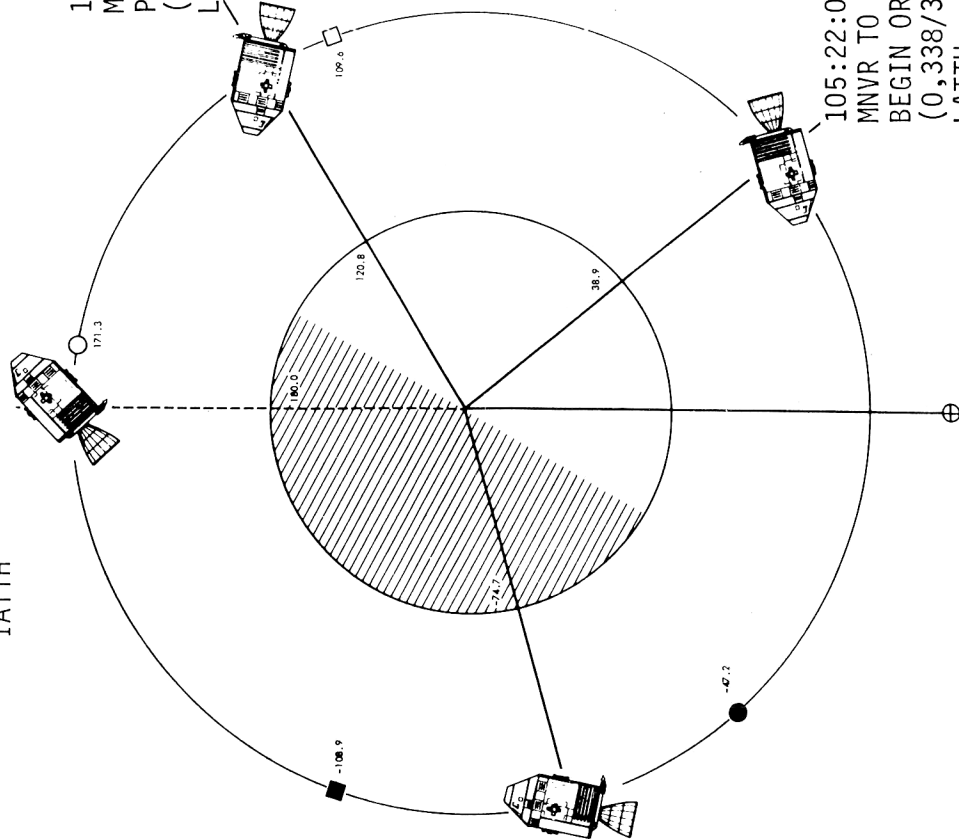
REVISION A

104:35:33
BEGIN REV 15
(320,25/133,348)
IATTH

104:55:00
MNVR TO ORB SCIENCE
PHOTO ATT
(0,230/279,0)
LATTH

105:22:00
MNVR TO LDMK TRKNG ATT;
BEGIN ORBRATE
(0,338/306,0)
LATTH

105:59:08
TERMINATE LATTH
IN SC CONTAMINATION
PHOTO ATT
(0,338/192,0)
IATTH



LEGEND:

MSFN AOS, LOS

S/C SUNRISE, SUNSET

SUBEARTH POINT

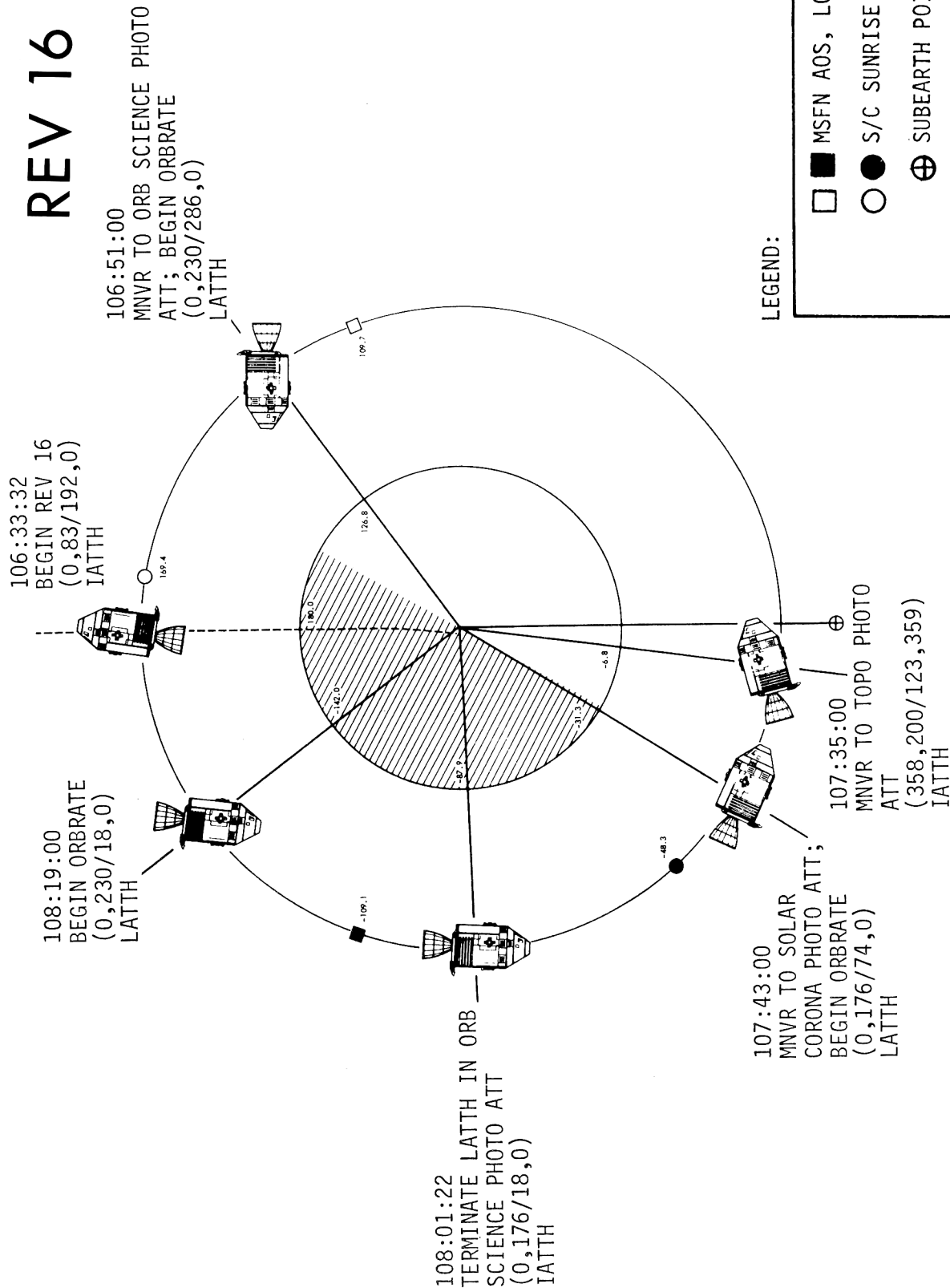
(R,LHP/INP,Y)

IATTH - INERTIAL ATTITUDE HOLD

LATTH - LOCAL ATTITUDE HOLD

3-78B

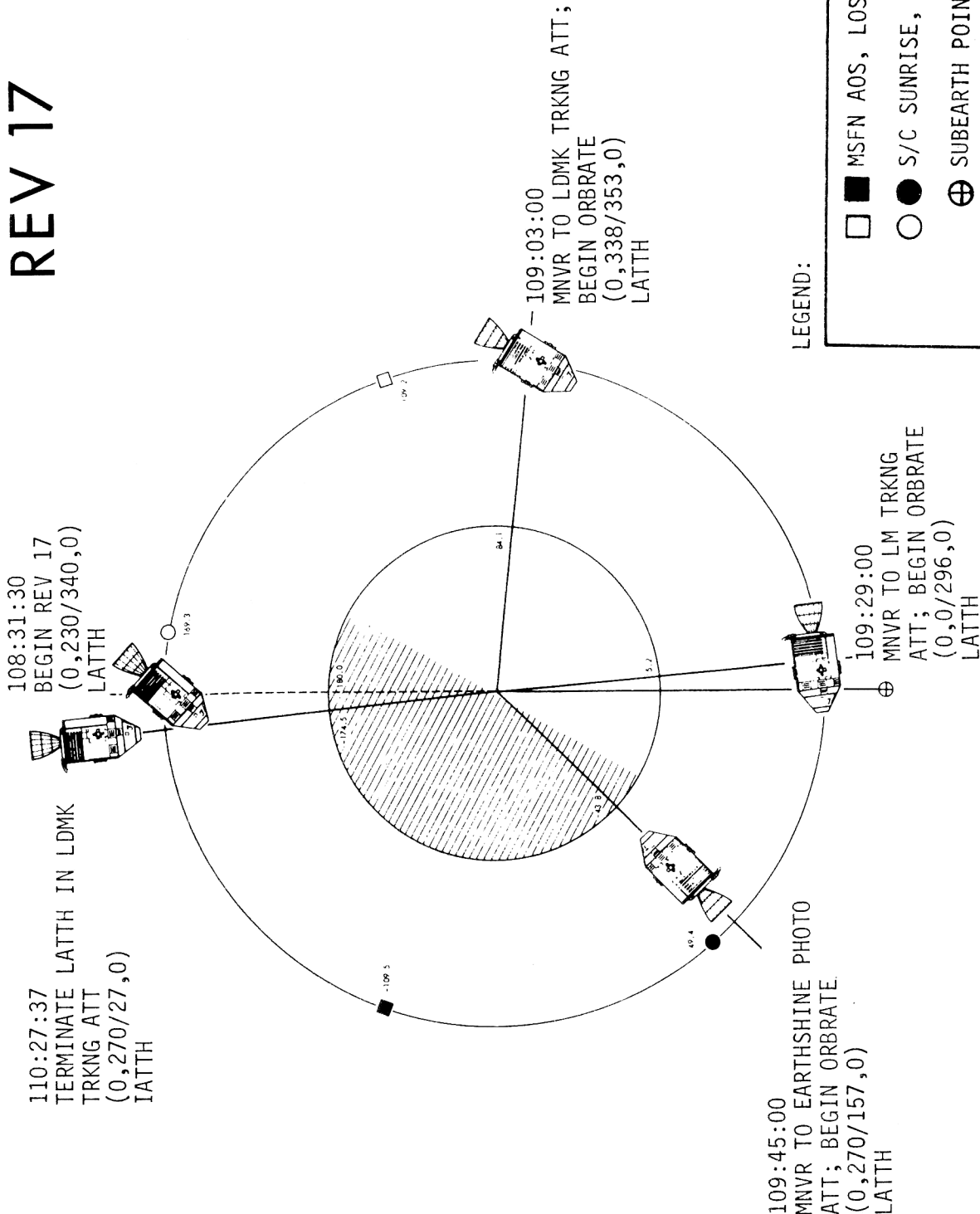
REV 16



3-80C

REVISION A

REV 17

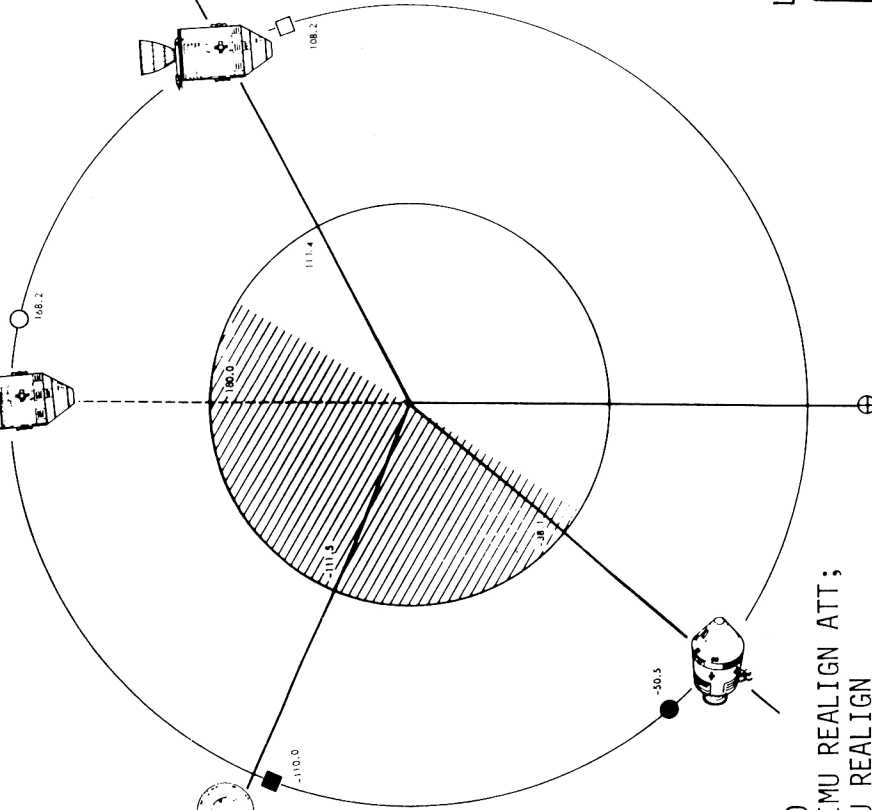


3-82C

REVISION A



111:41:00
MNVR TO IMU REALIGN ATT;
BEGIN IMU REALIGN
(0,204/115,45)
IATTH
AFTER ALIGN
(57,NA/34,334)



☐ MSFN AOS, LOS

○ ● S/C SUNRISE, SUNSET

⊕ SUBEARTH POINT

 $(R, LHP/INP, Y)$

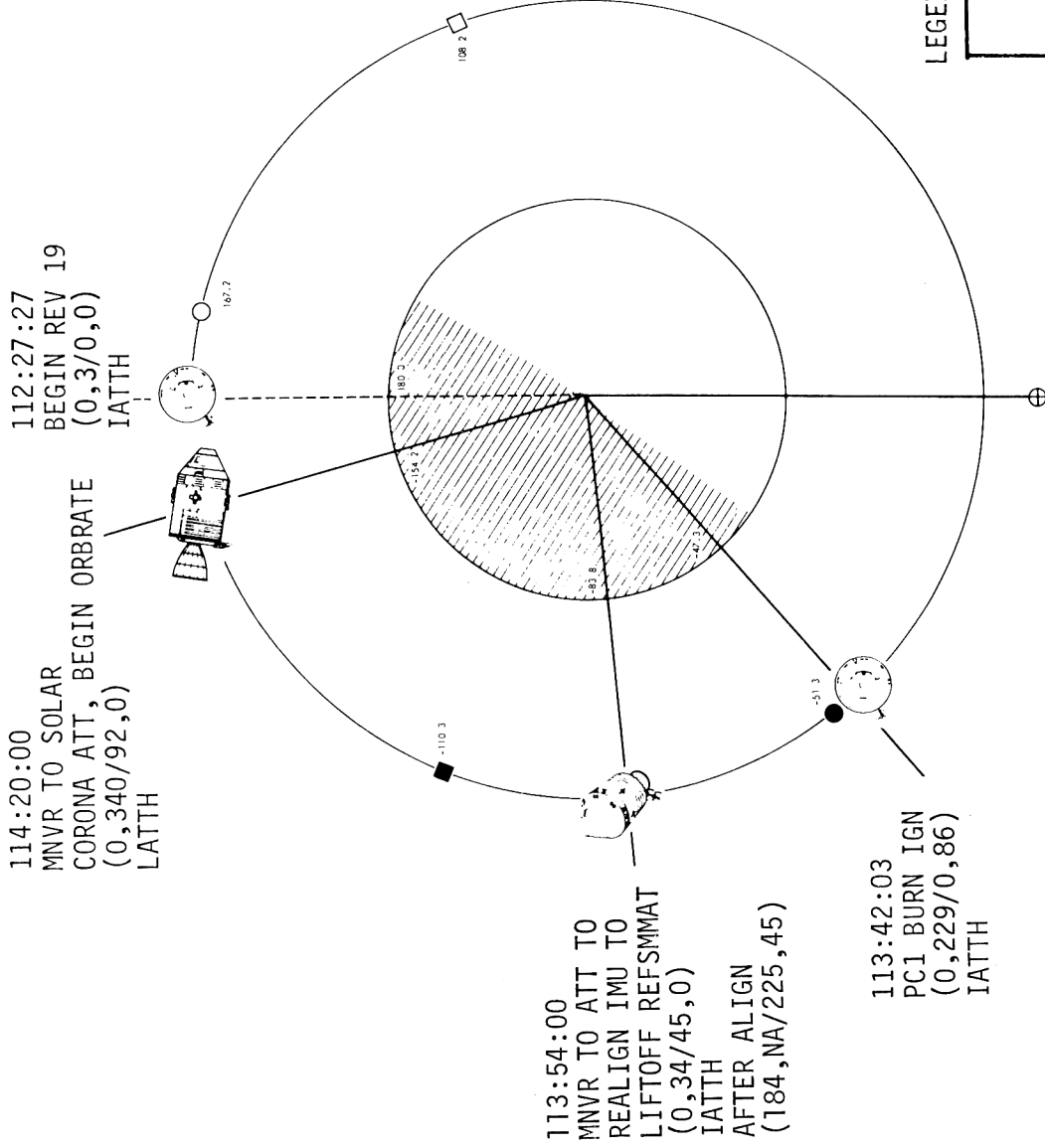
IATTH - INERTIAL ATTITUDE HOLD

LATTH - LOCAL ATTITUDE HOLD

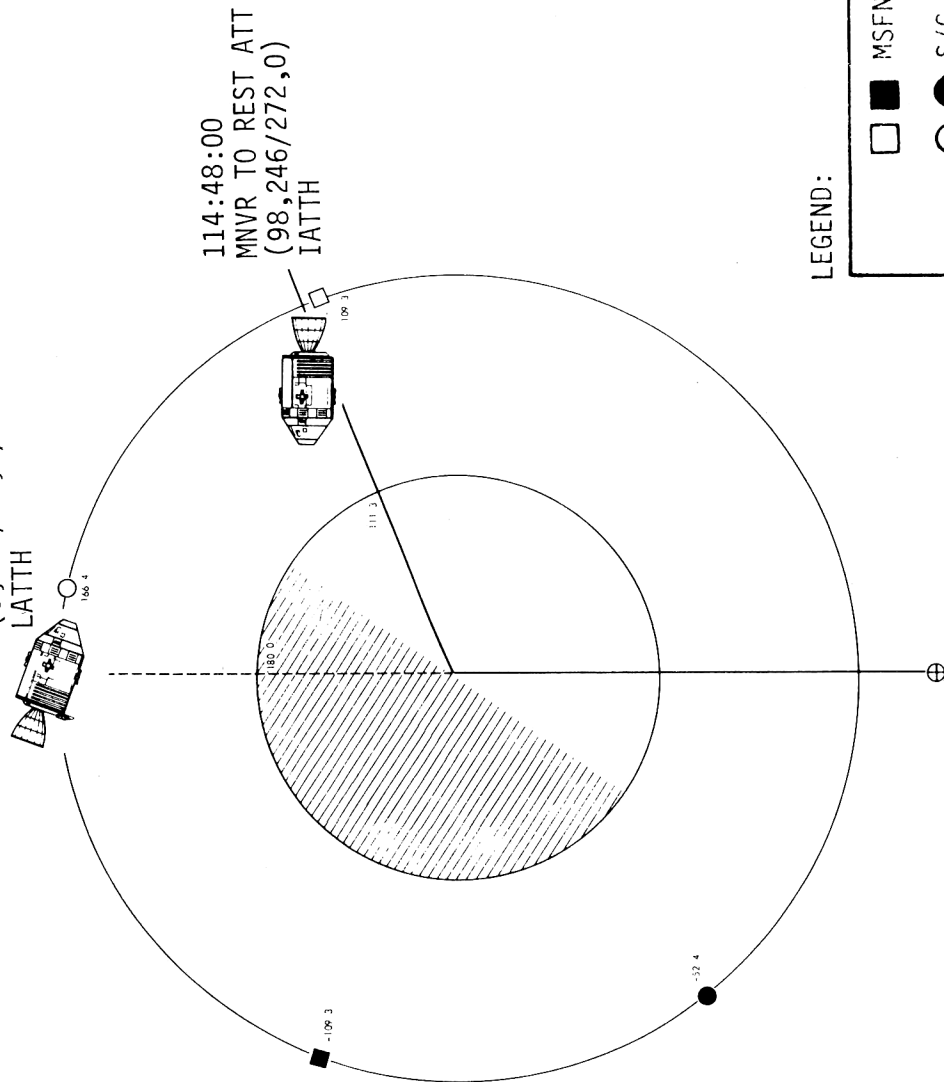
3-84B

REVISION A

REV 19



114:25:28
BEGIN REV 20
(0,340/75,0)



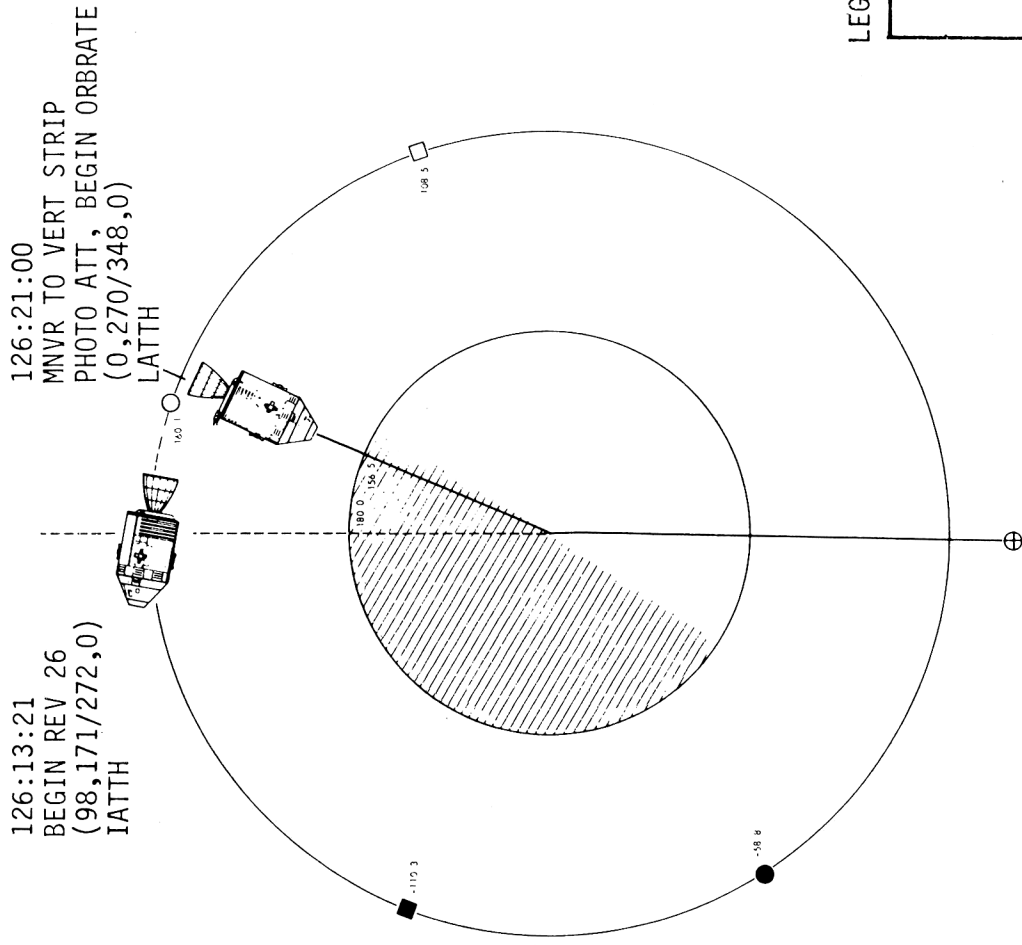
LEGEND:

- ☐ MSFN AOS, LOS
☒ S/C SUNRISE, SUNSET
☐ SUBEARTH POINT

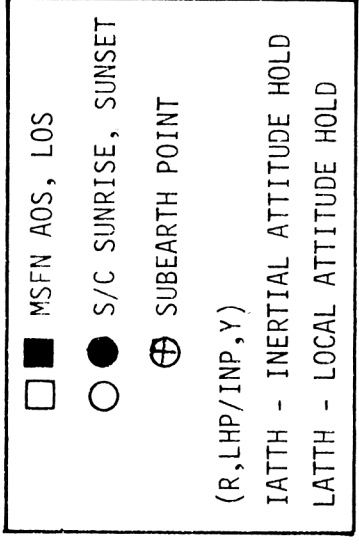
3-88A

REVISION A

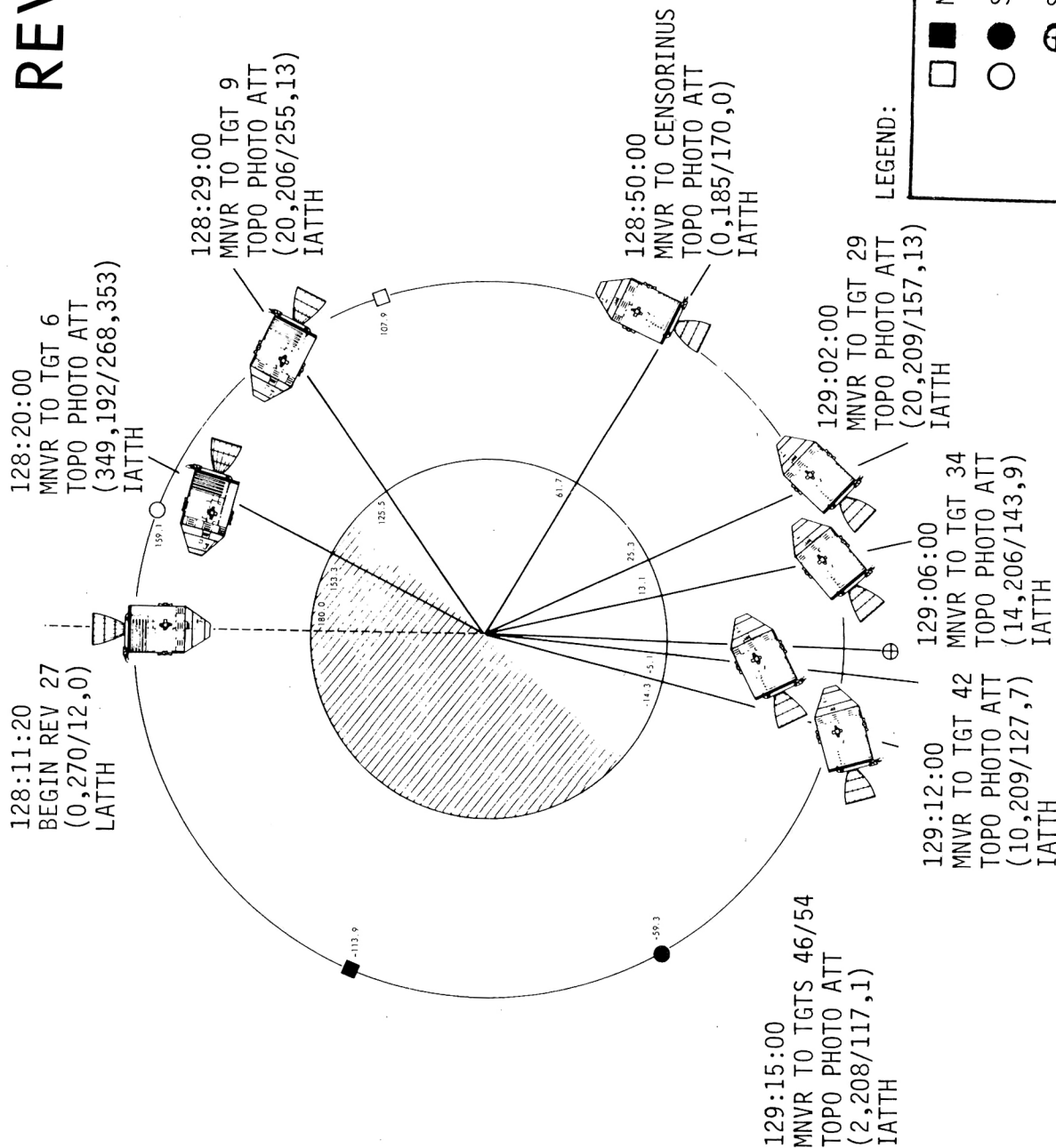
REV 26



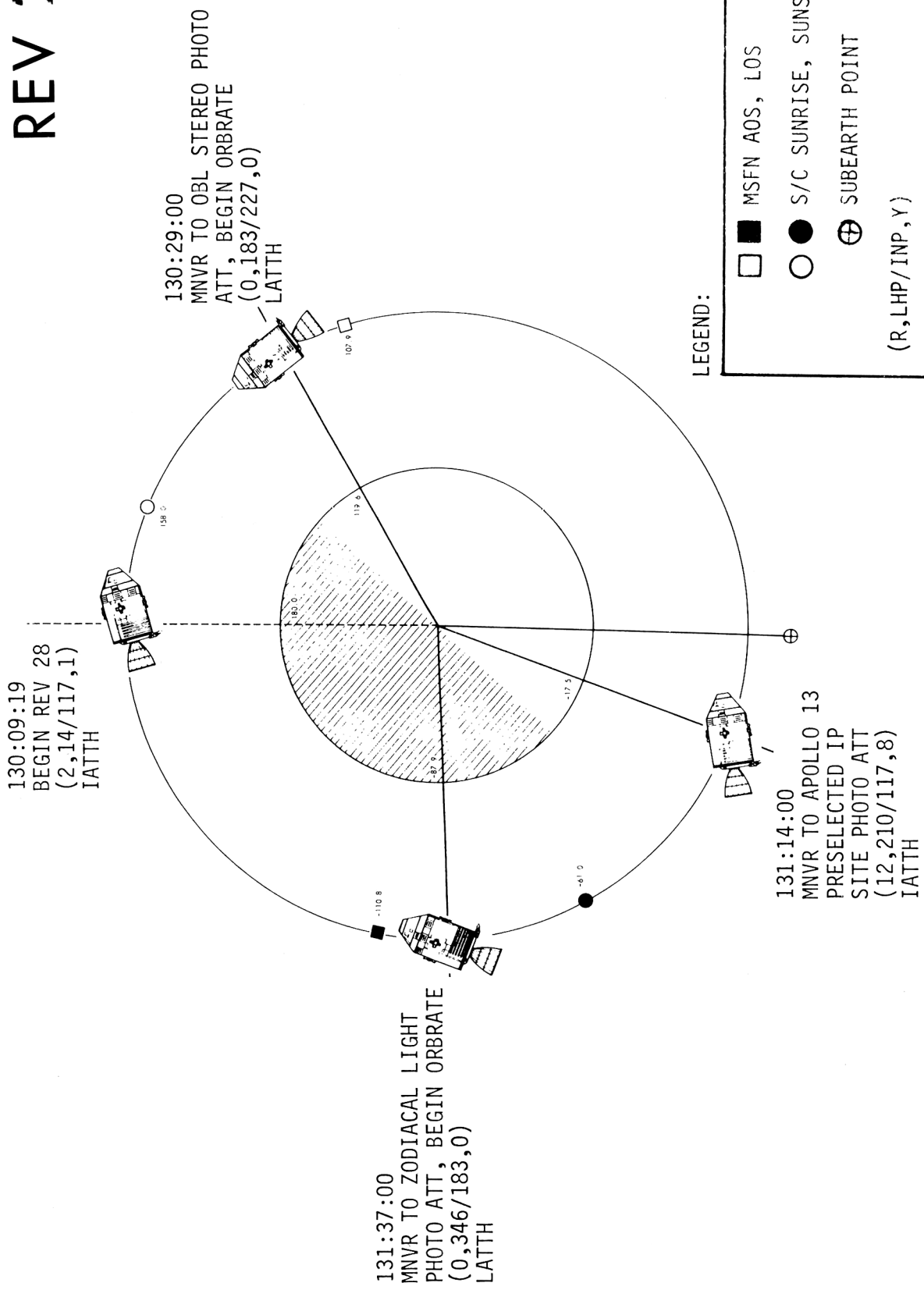
LEGEND:

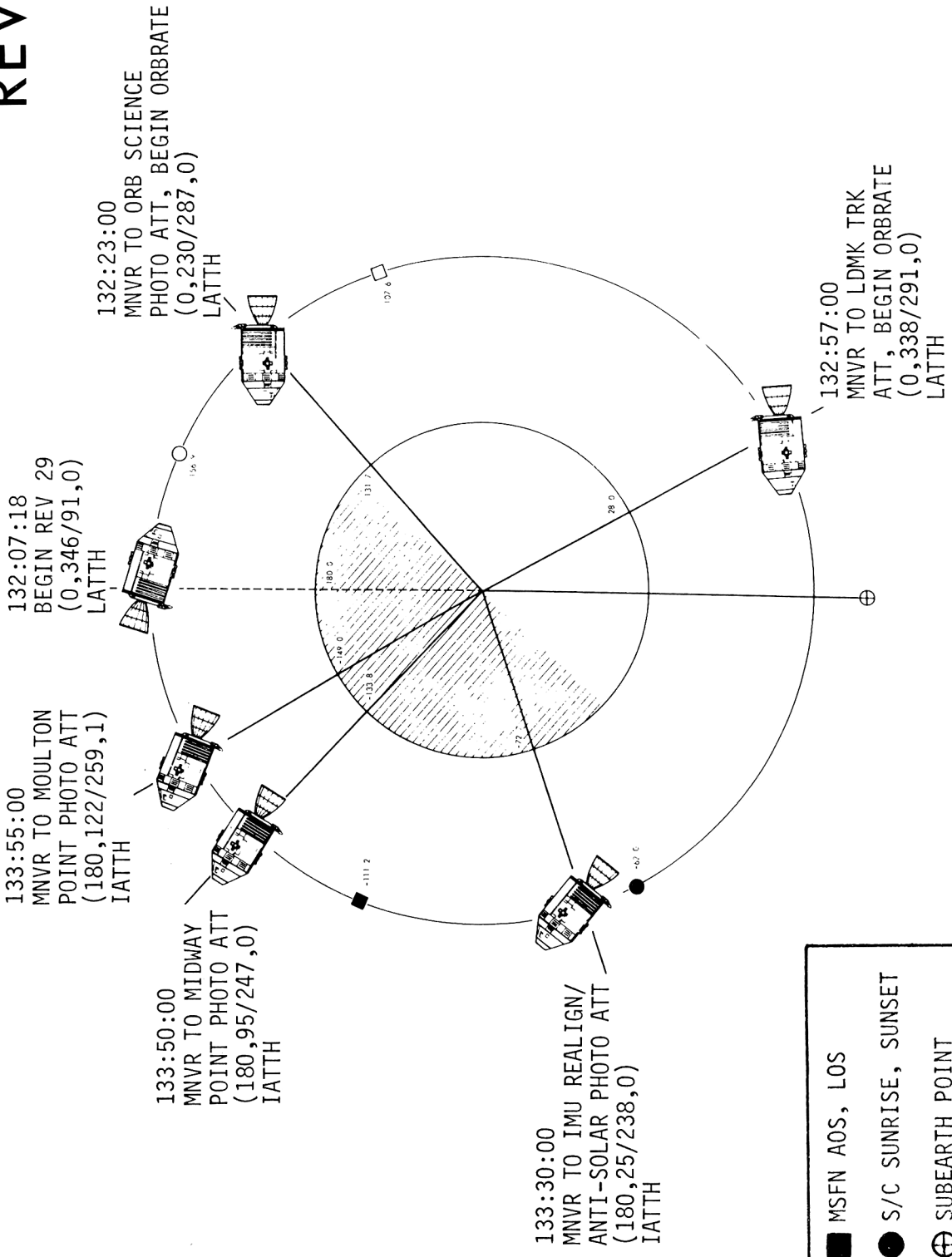


REVISION _____ A _____



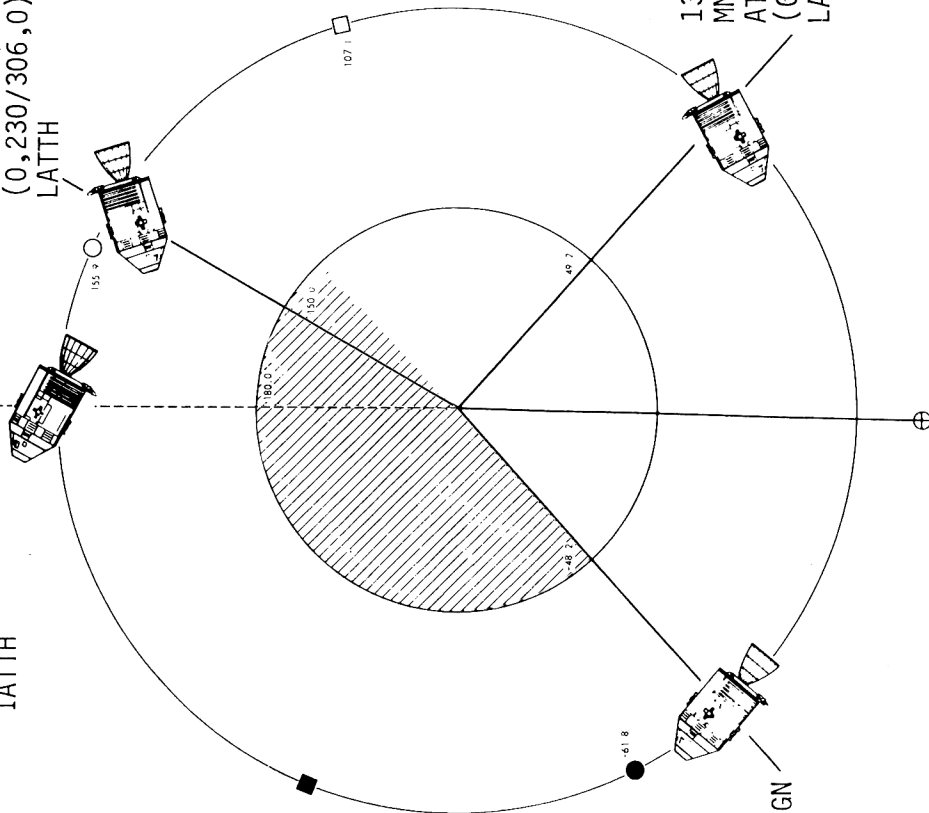
3-97B





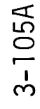
134:05:17
BEGIN REV 30
(179,153/259,1)
IATTH

134:15:00
MNVR TO ORB SCIENCE
PHOTO ATT, BEGIN ORBRATE
(0,230/306,0)
LATTH



135:20:00
MNVR TO SPICA,
BEGIN IMU REALIGN
AND COAS CAL
(180,1/240,358)
IATTH

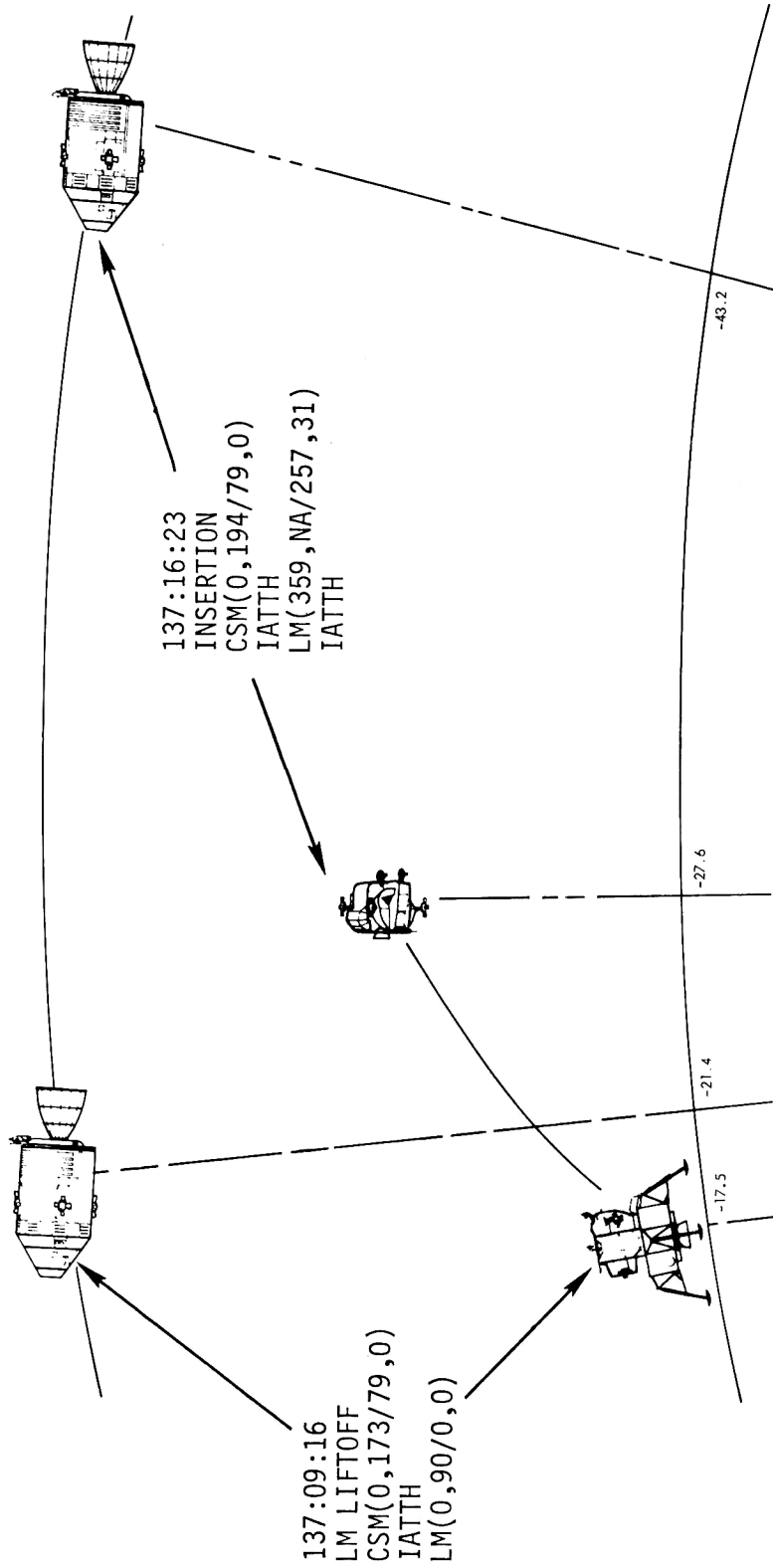
134:48:00
MNVR TO LDMK TRK
ATT, BEGIN ORBRATE
(0,338/314,0)
LATTH



☐ MSFN A0S, LOS
☒ S/C SUNRISE, SUNSET
☐ SUBEARTH POINT
 $\oplus$

(R,LHP/INP,Y).
 IATH - INERTIAL ATTITUDE HOLD
 LATH - LOCAL ATTITUDE HOLD

REVISION A



LEGEND:

	MSFN AOS, LOS
	S/C SUNRISE, SUNSET
	SUBEARTH POINT
(R,LHP/INF,Y)	
IATTH - INERTIAL ATTITUDE HOLD	
LATTH - LOCAL ATTITUDE HOLD	

3-107A

REVISION A

138:06:01
CSI BURN IGNITION
CSM(0,180/274,0)
IATTH
LM(0,359/191,0)
LOSM TO CSM

138:12:45
CSM AND LM BEGIN
VHF RNG AND RR TRKNG,
RESPECTIVELY
CSM(0,225/298,0)
LOSM TO LM
LM(0,2/173,0)
LOSM TO CSM

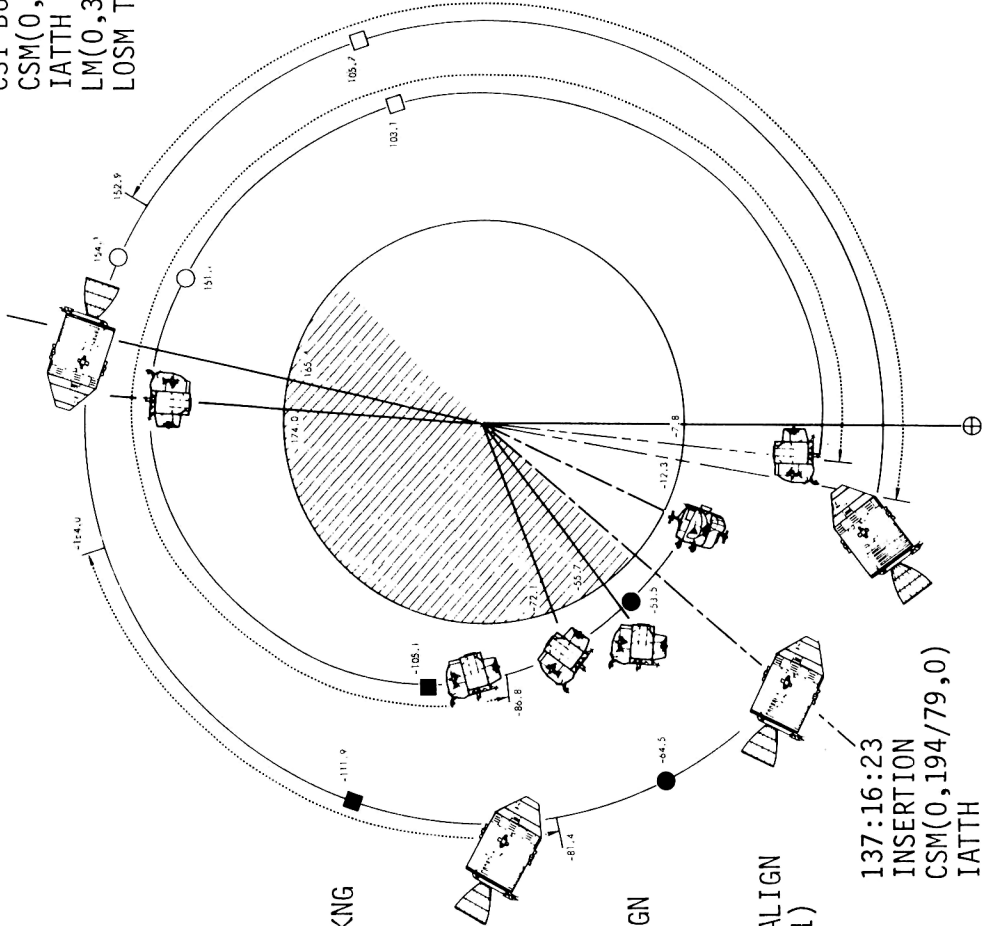
137:34:23
LM BEGIN RR TRKNG
LM(0,5/296,0)
LOSM TO CSM

137:28:53
CSM BEGIN VHF RNG
CSM(0,232/79,0)
LOSM TO LM

137:29:53
LM BEGIN IMU REALIGN
LM(359,25/331,1)
IATTH

137:24:53
LM BEGIN IMU REALIGN
LM(359,323/285,1)
IATTH

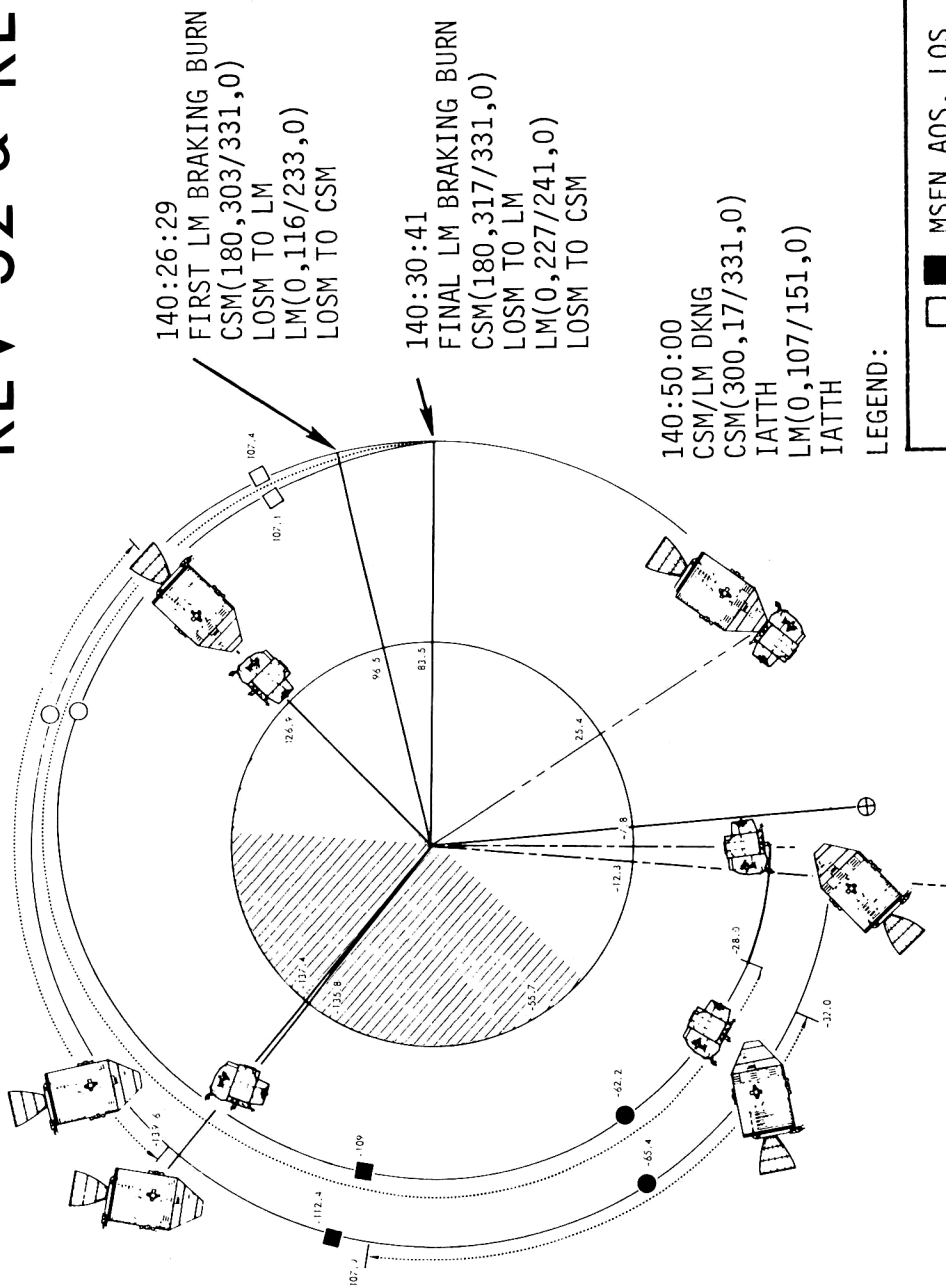
137:16:23
INSERTION
CSM(0,194/79,0)
IATTH
LM(359,NA/257,31)
IATTH



REVISION A

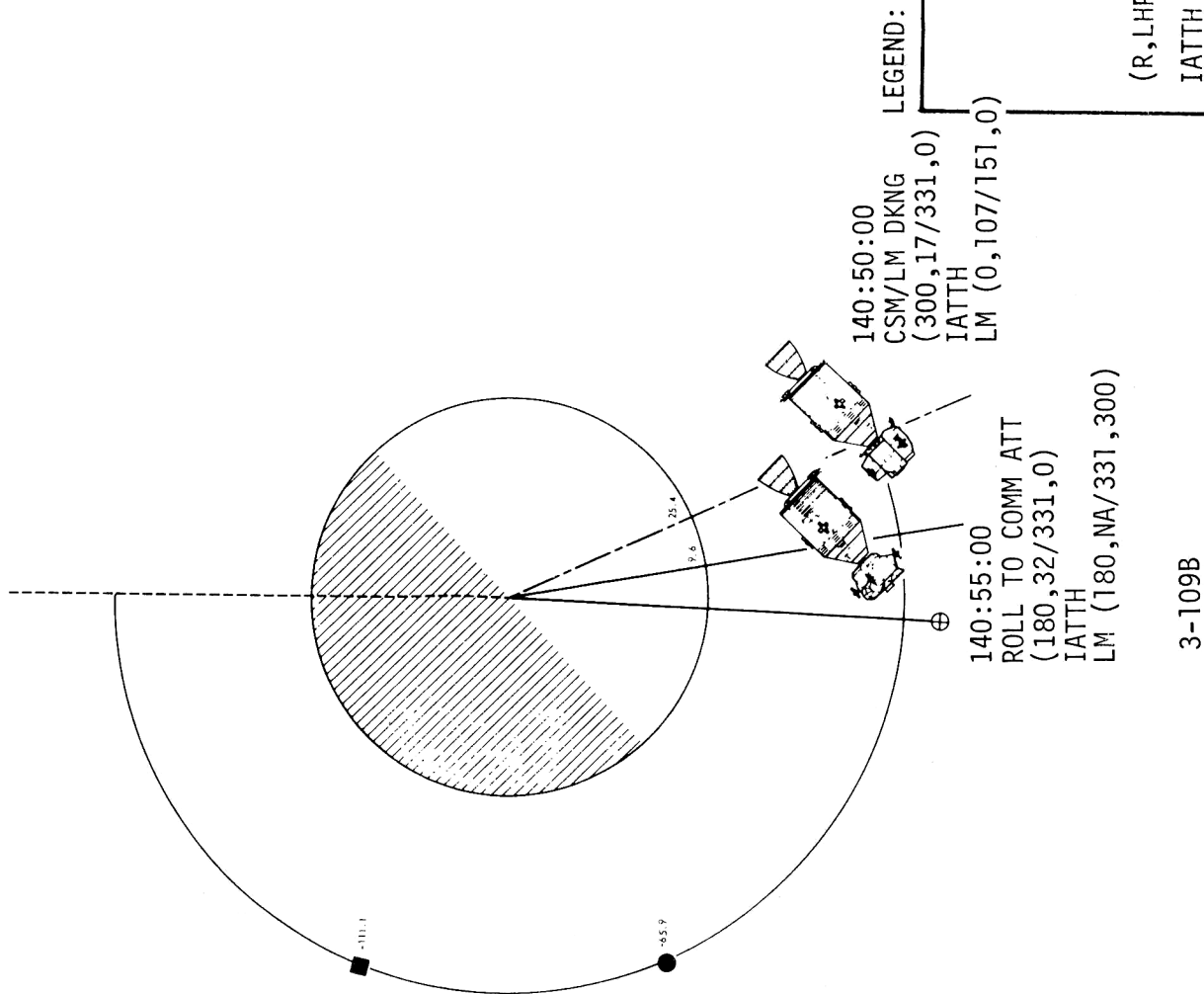
REV 32 & REV 33

139:45:14
TPI BURN IGNITION
CSM(0,208/0,0)
IATTH
LM(0,26/270,0)
LOSM TO CSM



3-109A

REVISION A



LEGEND:

□	MSFN AOS, LOS
●	S/C SUNRISE, SUNSET
⊕	SUBEARTH POINT
(R,LHP/INP,Y)	
IATTH - INERTIAL ATTITUDE HOLD	
LATTH - LOCAL ATTITUDE HOLD	

REVISION _____ A

3-109B

CSM

CMP

MCC-1 BACKUP
CONFIRM LM MCC-1
P35-TARGET MCC-2
SXT & VHF TRACKING

FINAL MCC-2 COMP
P41-RCS THRUSTING
MCC-2 BACKUP
CONFIRM LM MCC-2
PREDOCK CHECKLIST
P47-THRUST MONITOR
ROLL 180° FOR HI GAIN
HGA P -47, Y 358
TV (MAD) 140:23-140:35
START DAC

PITCH 360° FOR LM PHOTOS

GO/NO-GO FOR PYRO ARM
(CUE MSFN), LOGIC-ON
ROLL RT 120° TO DOCK ATT:
R 300, P 332, Y 0

OMNI D
PYRO ARM

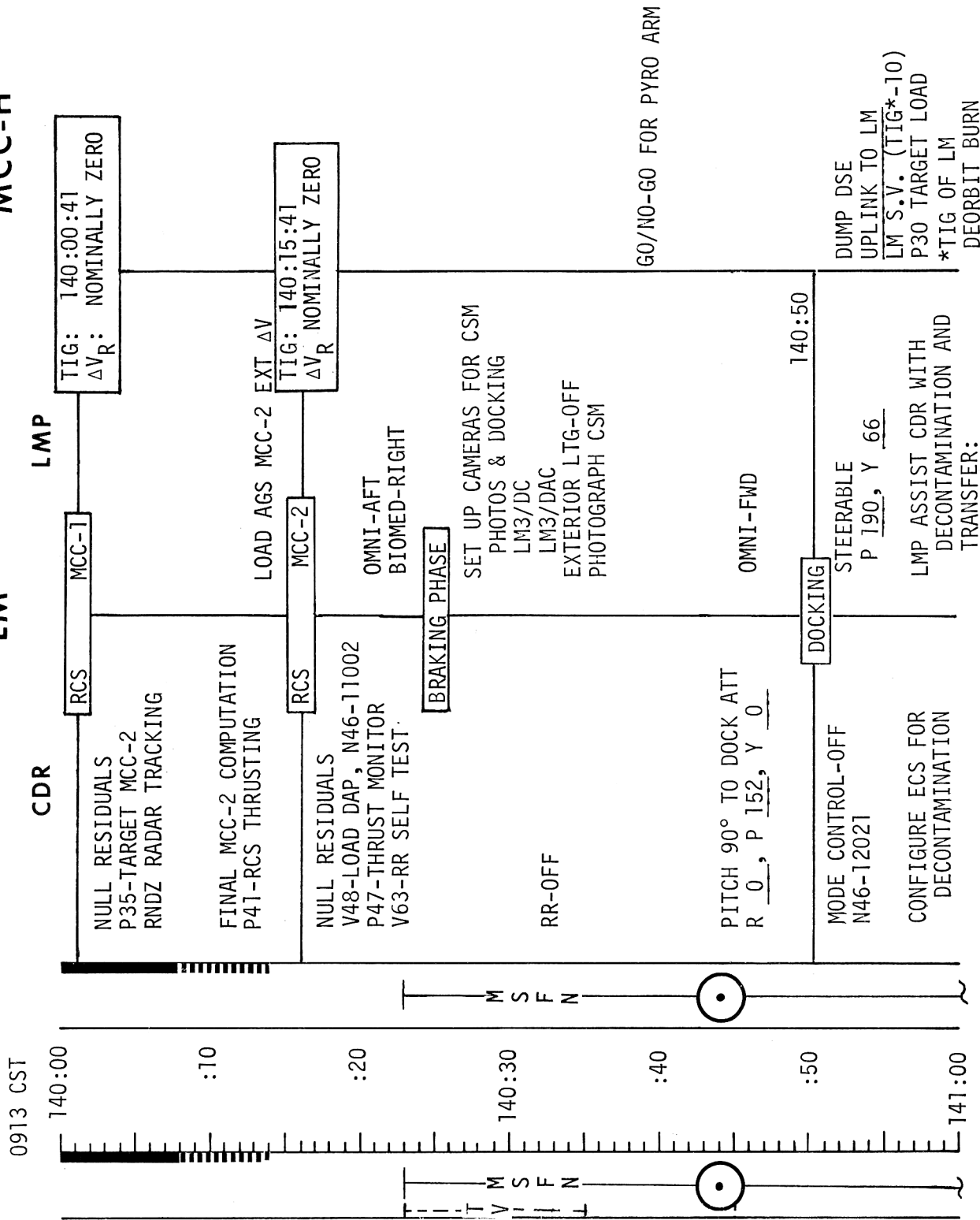
CSM ACTIVE DOCKING

ROLL LEFT 120° TO R 180
HGA: P -45, Y 356
SC CONT-SCS
N46 (61112, 11111)

FLIGHT PLAN

LM

MCC-H



MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 13	FINAL (APRIL)	MARCH 27, 1970	140:00 - 141:00	6/33	3-110

FLIGHT PLANNING BRANCH

REVISION A

MCC-H

1013 CST

✕

LM

DISCONNECT & STOW VACUUM
BRUSH & HOSES

```
VERIFY TUNNEL PRESSURIZED
OPEN HATCH
RECEIVE PROBE FROM CMP & STOW
RECEIVE DROGUE FROM CMP & STOW
OVER PROBE
```

OVER 1 MODE
RECEIVE BAGS FROM CSM
BAG AND TRANSFER TO CMP:

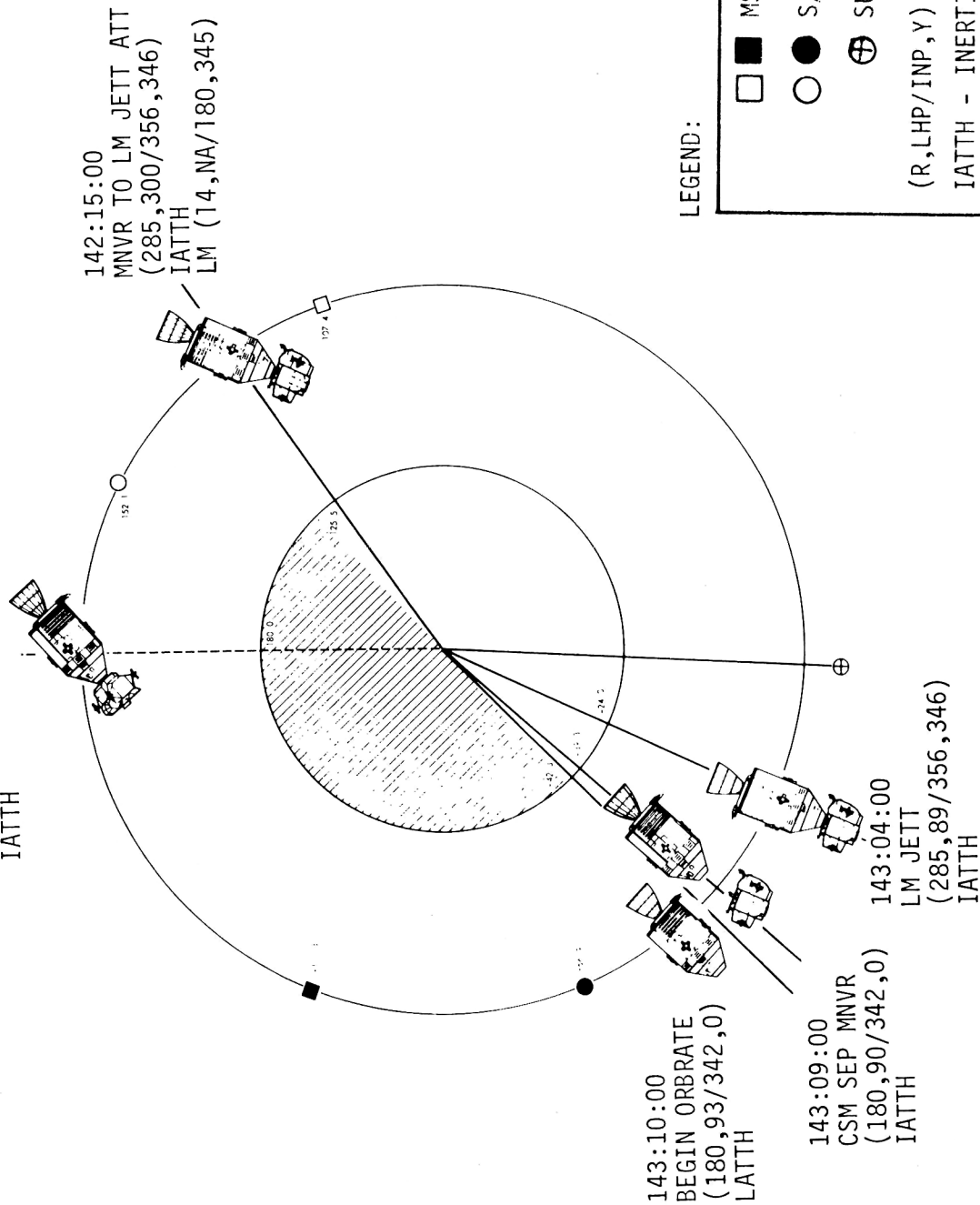
1 ISA W/ TOTE BAG & WEIGH BAG
2 HELMETS (GLOVES INSIDE)
2 SRC'S FROM SRC RACKS
1 TOTE BAG W/DUST COVER CONTAINING

—5	70 MM	MAGS	IN BAGS (3+2)
—6	16 MM	MAGS	IN BAG
—1	70 MM	CAMERA	
—1	CSC	CASSETTE	
—1	CLSRC		
—2	16 MM	MAGS	IN BAG

MCC VERIFY DSE
MOTION BEFORE
LOS

MSC Form 29 (May 69)

FLIGHT PLANNING BRANCH



LEGEND:

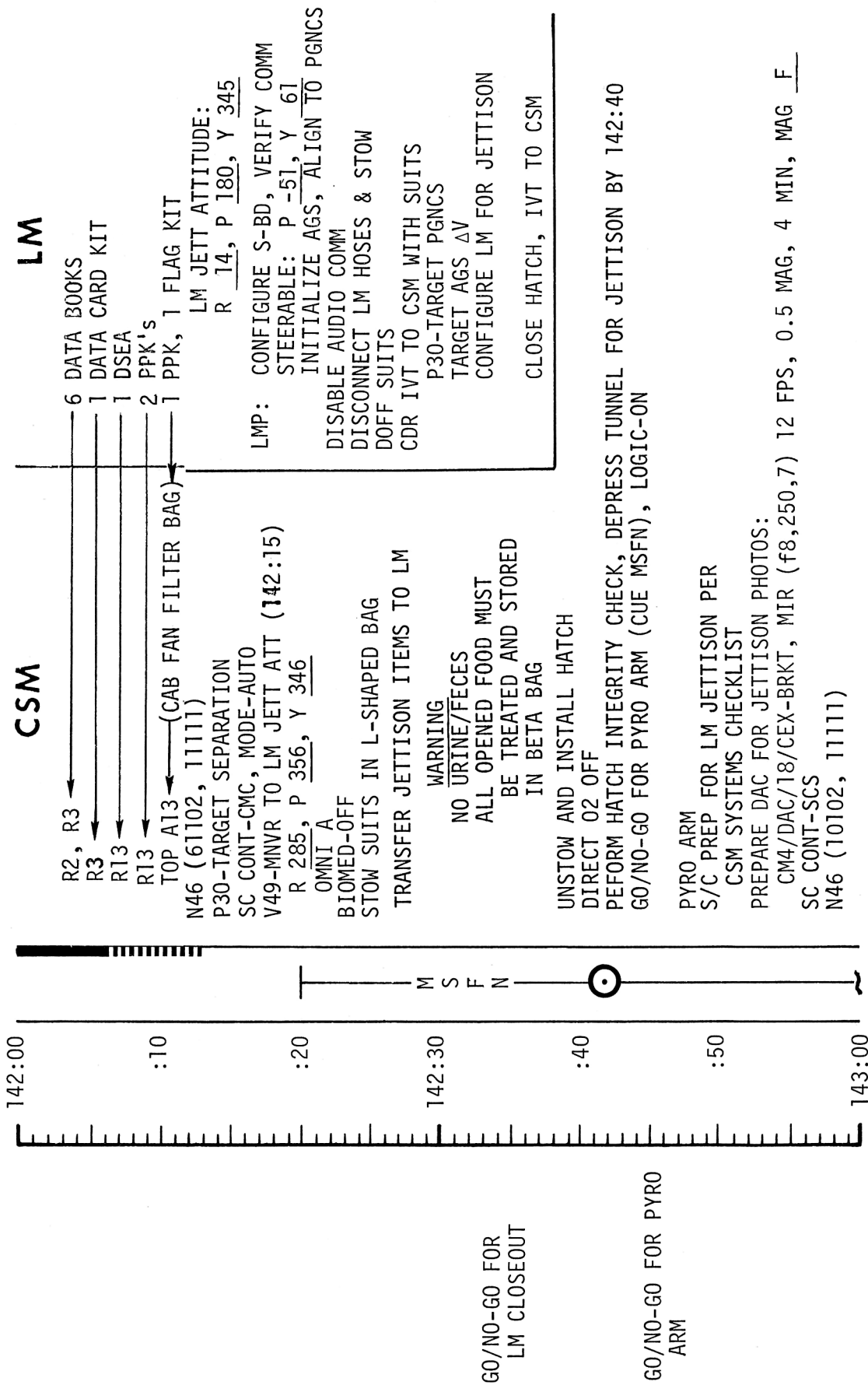
- ☐ MSFN AOS, LOS
☐ S/C SUNRISE, SUNSET
☒ SUBEARTH POINT
- (R,LHP/INP,Y)
- IATTH - INERTIAL ATTITUDE HOLD
 LATTH - LOCAL ATTITUDE HOLD

REVISION A

MCC-H

1113 CST

FLIGHT PLAN



MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 13	FINAL (APRIL)	MARCH 27, 1970	142:00 - 143:00	6/34	3-112

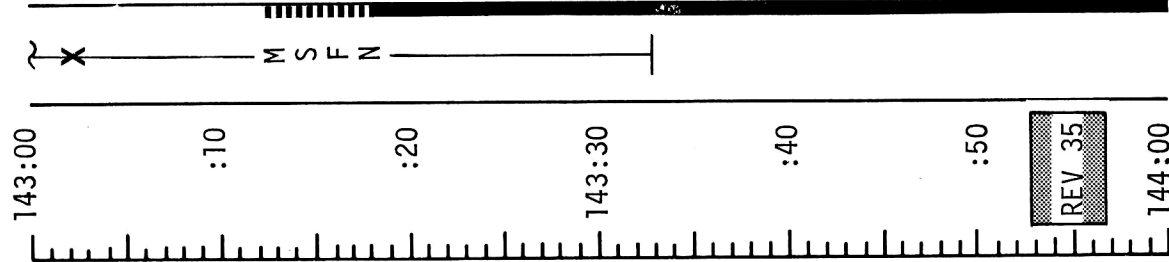
MSC Form 29 (May 69)

FLIGHT PLANNING BRANCH

REVISION A

FLIGHT PLAN

1213 CST



DUMP DSE
UPLINK TO LM
V48-LOAD DAP
R1-12011
P42-APS THRUSTING
UPDATE TO CSM
TOPO PHOTO PAD REV 35

DAC ON
P47-THRUST MONITOR
LM JETTISON 143:04
P41-RCS THRUSTING
MNVR TO SEP ATT, (143:08) R 180, P 341.6, Y 0
OMNI B

CSM SEPARATION 143:09

GO ORB RATE (143:10) P 093/342, V64-ACQ MSFN
PREPARE DAC FOR LM DEORBIT BURN PHOTOGRAPHY
CM/DAC/SXT/CEX (FIXED,250,FIXED) 12 FPS, 0.5 MAG, 4 MIN
MAG F

S/C CLEANUP

CMP DOFF PGA
CONFIGURE FOR BI-STATIC RADAR TEST
VERIFY VHF AM B - DUPLEX
VHF RANGING-RANGING
VHF ANTENNA-LEFT
VHF AM SQUELCH A-MAX
VHF AM T/R OFF

VERIFY DSE MOTION @ LOS

(3 AUDIO PANELS)
THIS VHF CONFIGURATION WILL BE
MAINTAINED UNTIL 166:10

EAT PERIOD

TOPO PHOTO PAD REV 35
R P Y
T START: : :
T STOP : : :
RNG NM.

TIG: 143:09
BT: 6.12 SEC
ΔVR: 1.0 FT/SEC
ULLAGE: N/A
ORBIT: 57.7 x 55.3

LM IS TARGETED FOR
APS IMPULSE BURN.
THRUST IS RCS
ULLAGE ONLY.

ONBOARD READOUT

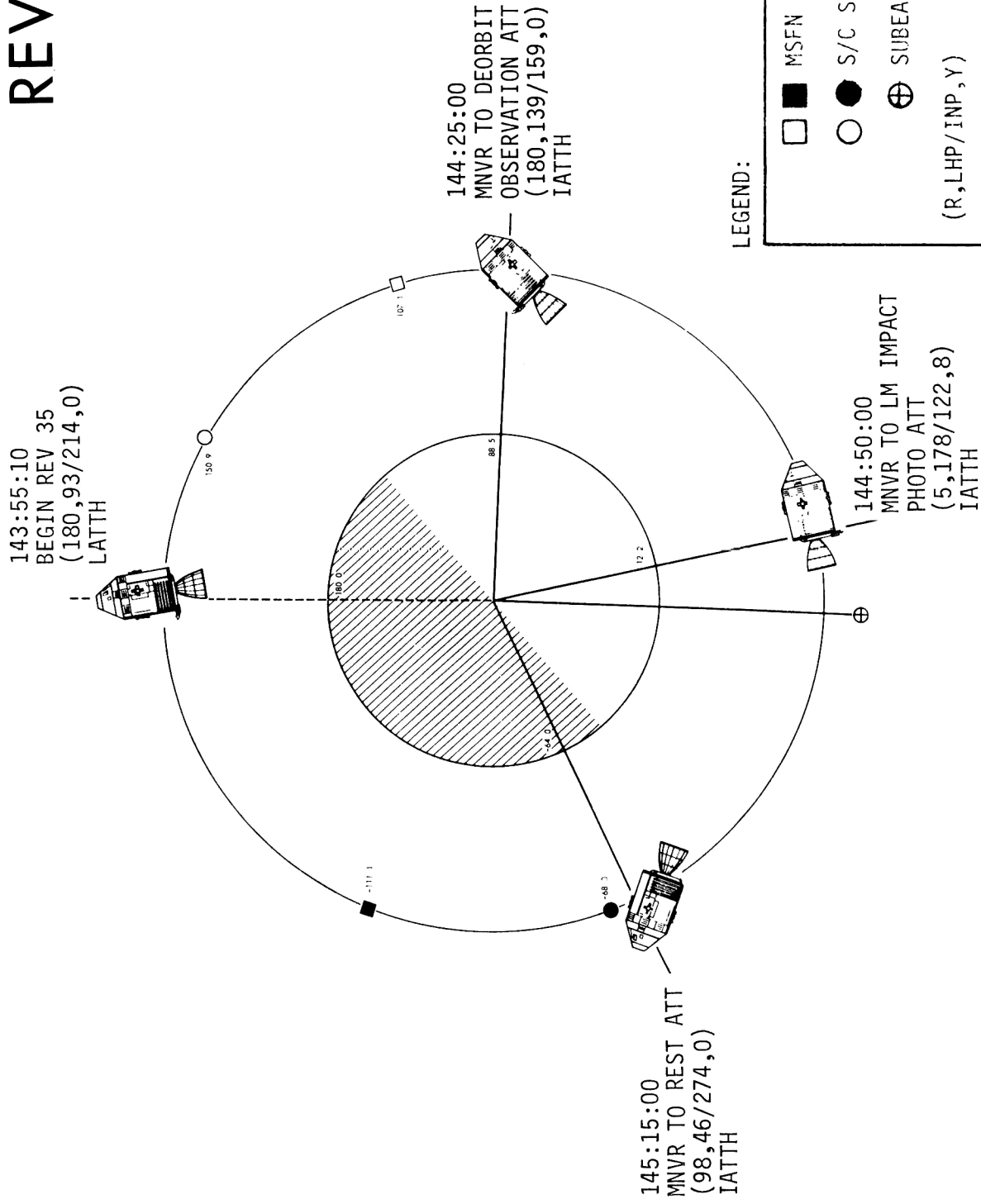
BAT C
PYRO BAT A
PYRO BAT B
RCS A
B
C
D

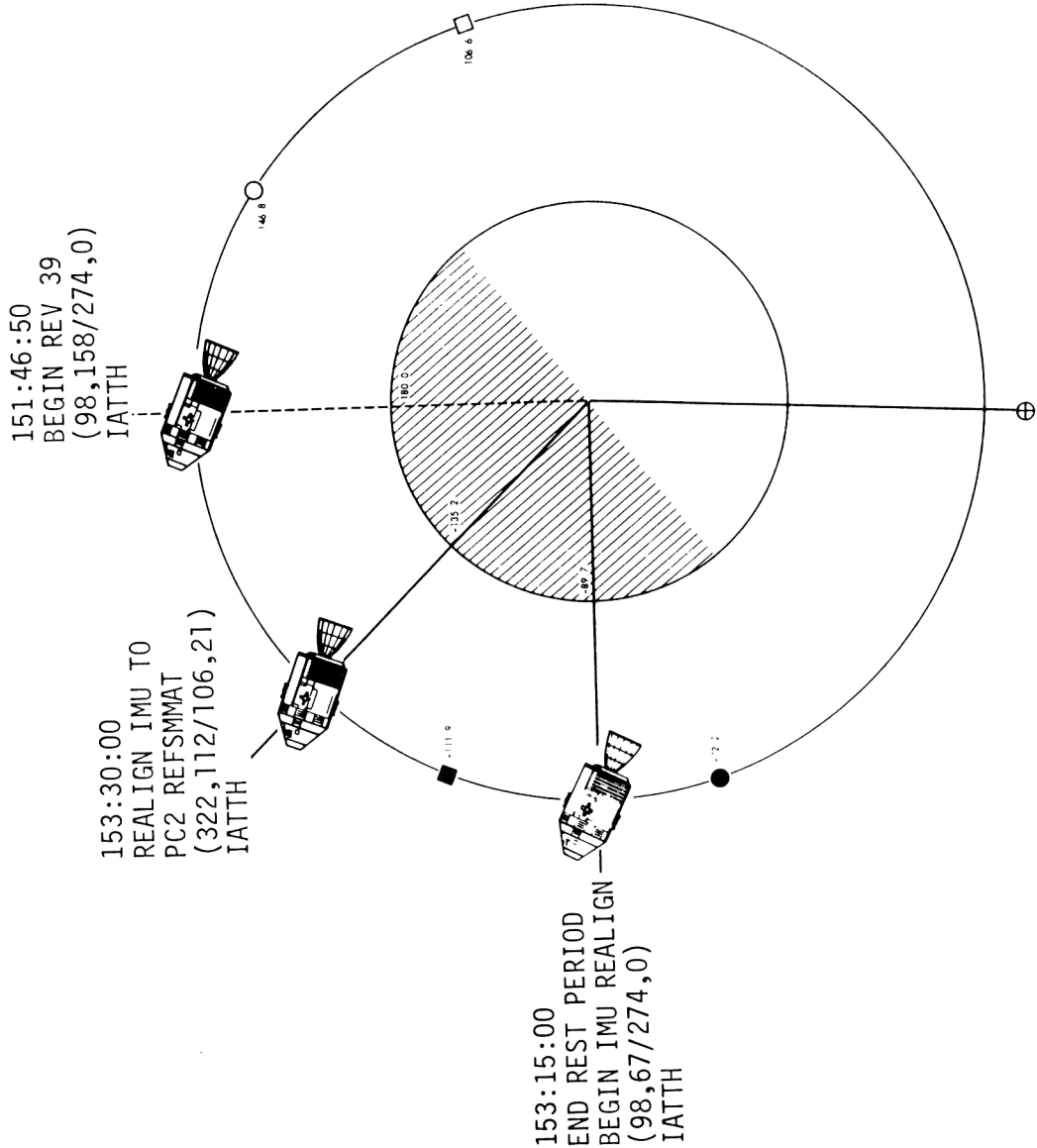
DC IND SEL - MNA OR B

MAP UPDATE REV 35

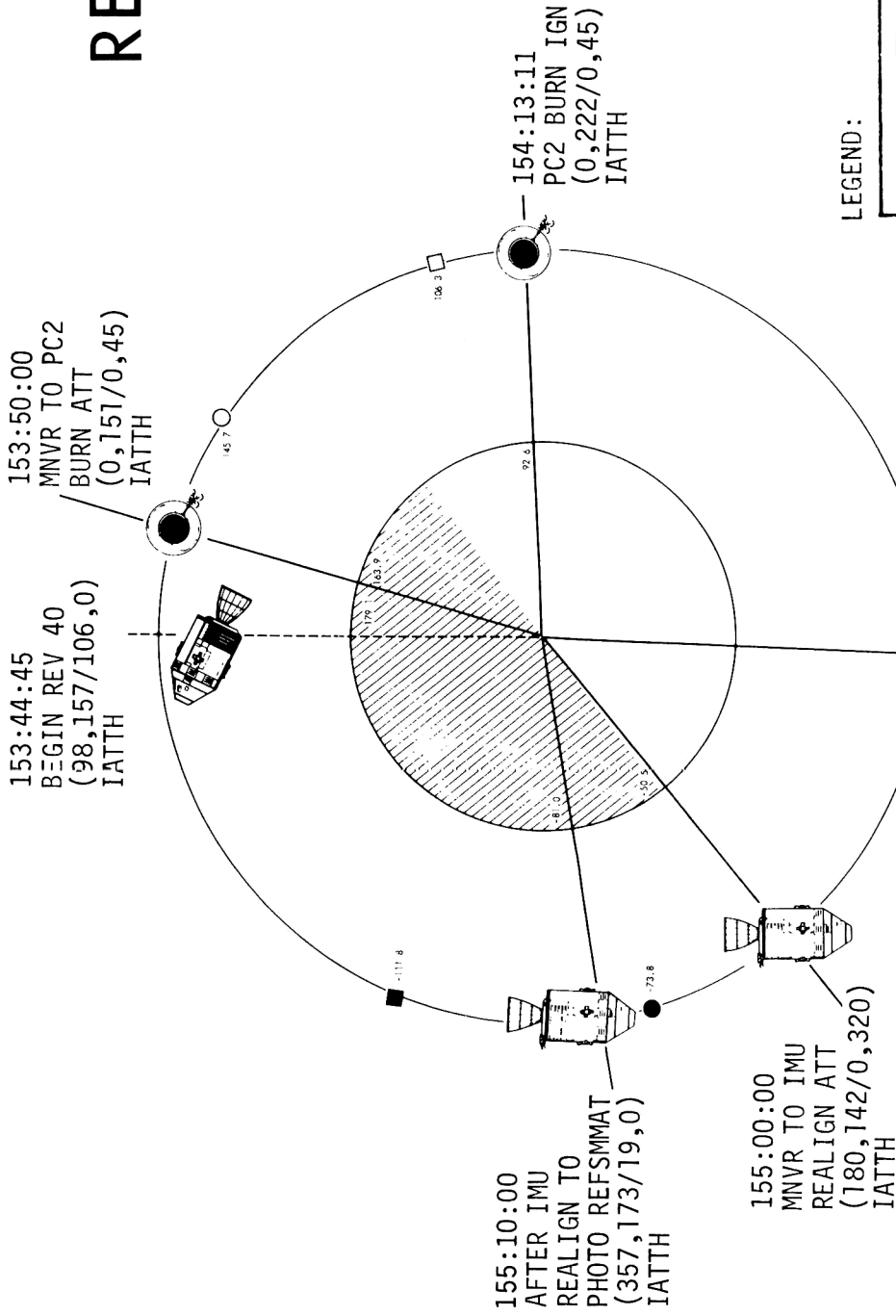
LOS : : :
180° : : :
AOS : : :

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 13	FINAL (APRIL)	MARCH 27, 1970	143:00 - 144:00	6/34-35	3-113





REV 40

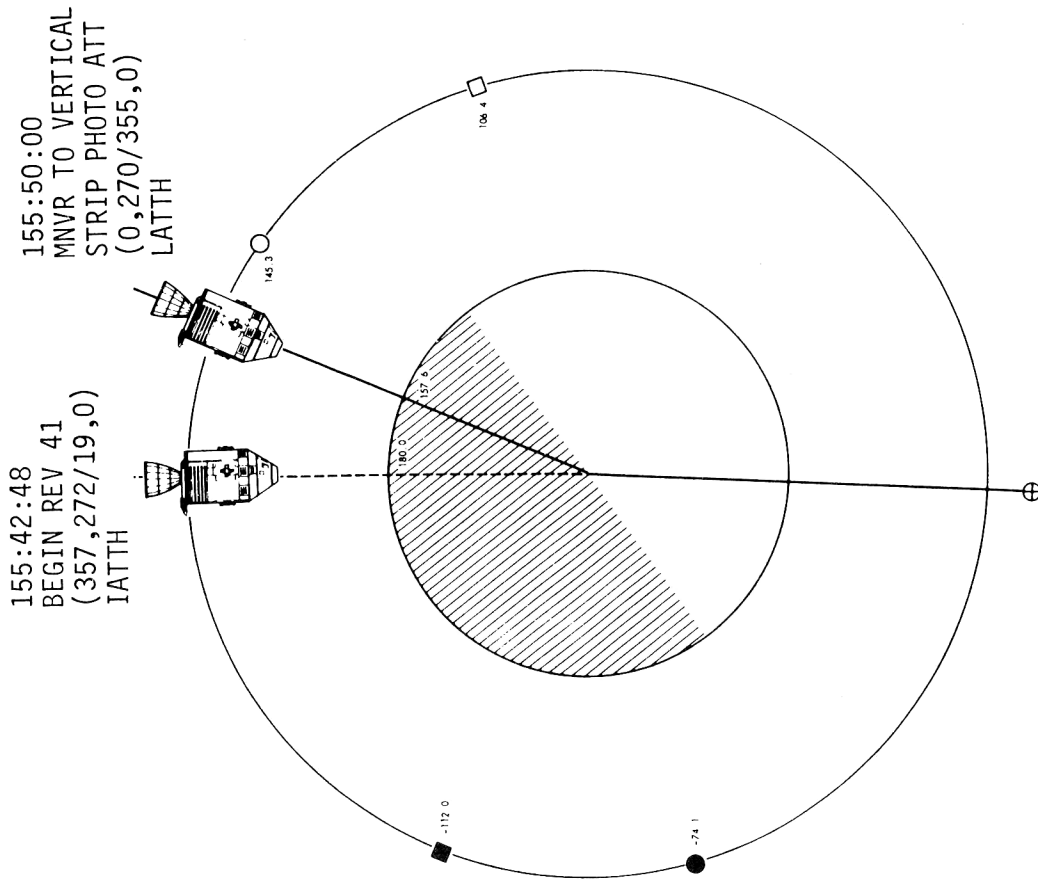


LEGEND:

□	MSFN AOS, LOS
○	S/C SUNRISE, SUNSET
⊕	SUBEARTH POINT
(R,LHP/INP,Y)	
IATTH	- INERTIAL ATTITUDE HOLD
LATTH	- LOCAL ATTITUDE HOLD

REVISION A

REV 41

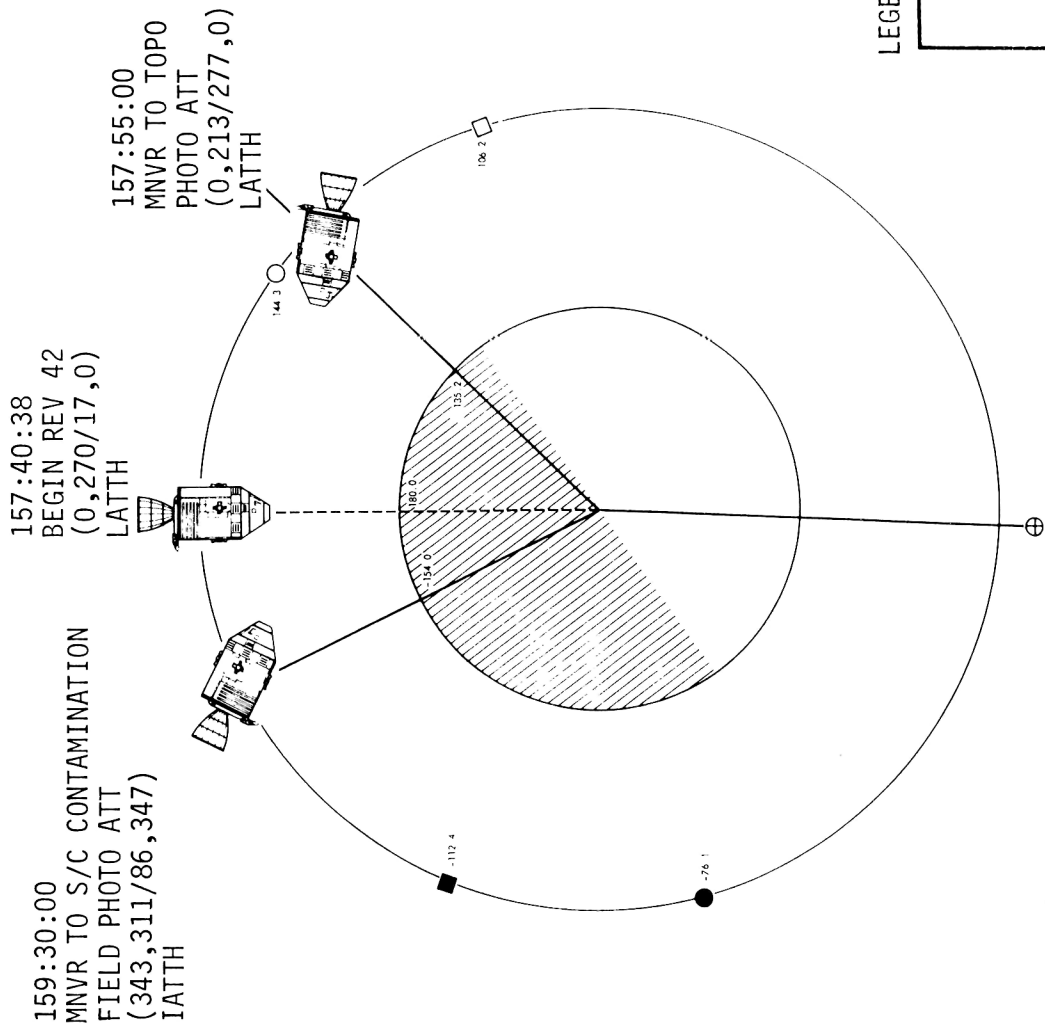


LEGEND:

	MSFN AOS, LOS
	S/C SUNRISE, SUNSET
	SUBEARTH POINT
(R, LHP/INP, Y)	
IATTH - INERTIAL ATTITUDE HOLD	
LATTH - LOCAL ATTITUDE HOLD	

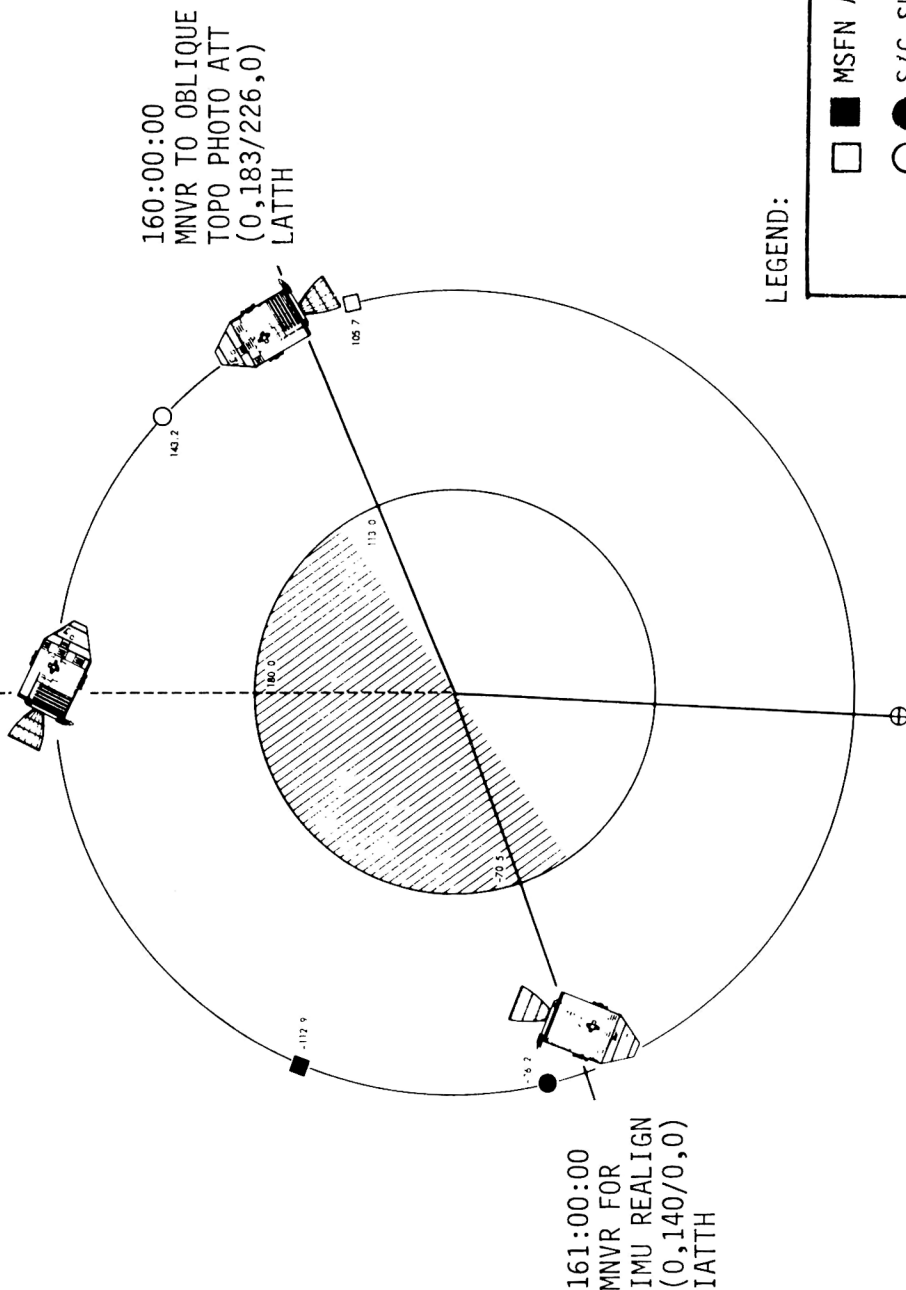
REVISION A

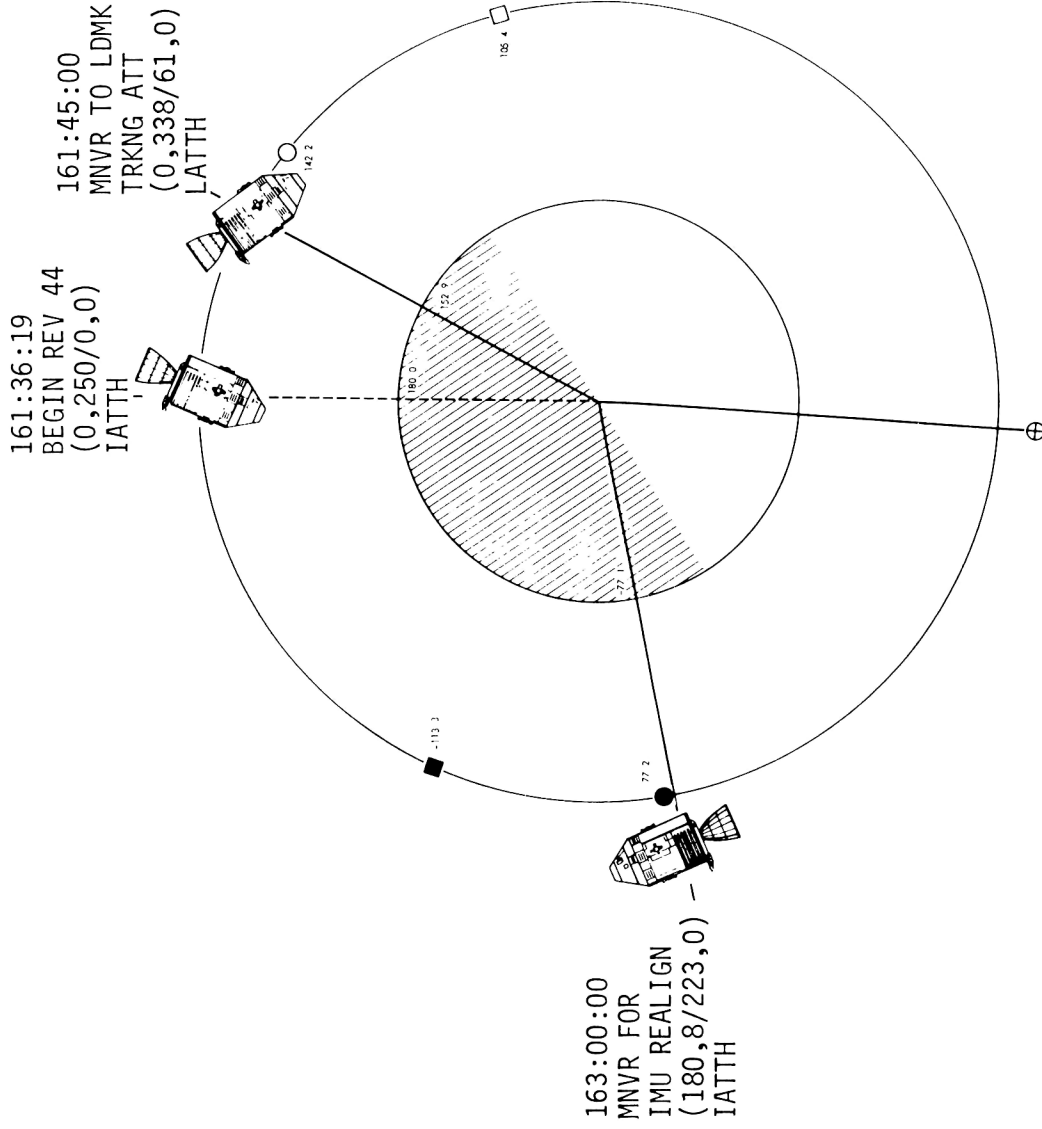
REV 42



159:38:28
 BEGIN REV 43
 (343,337/86,347)
 IATTH

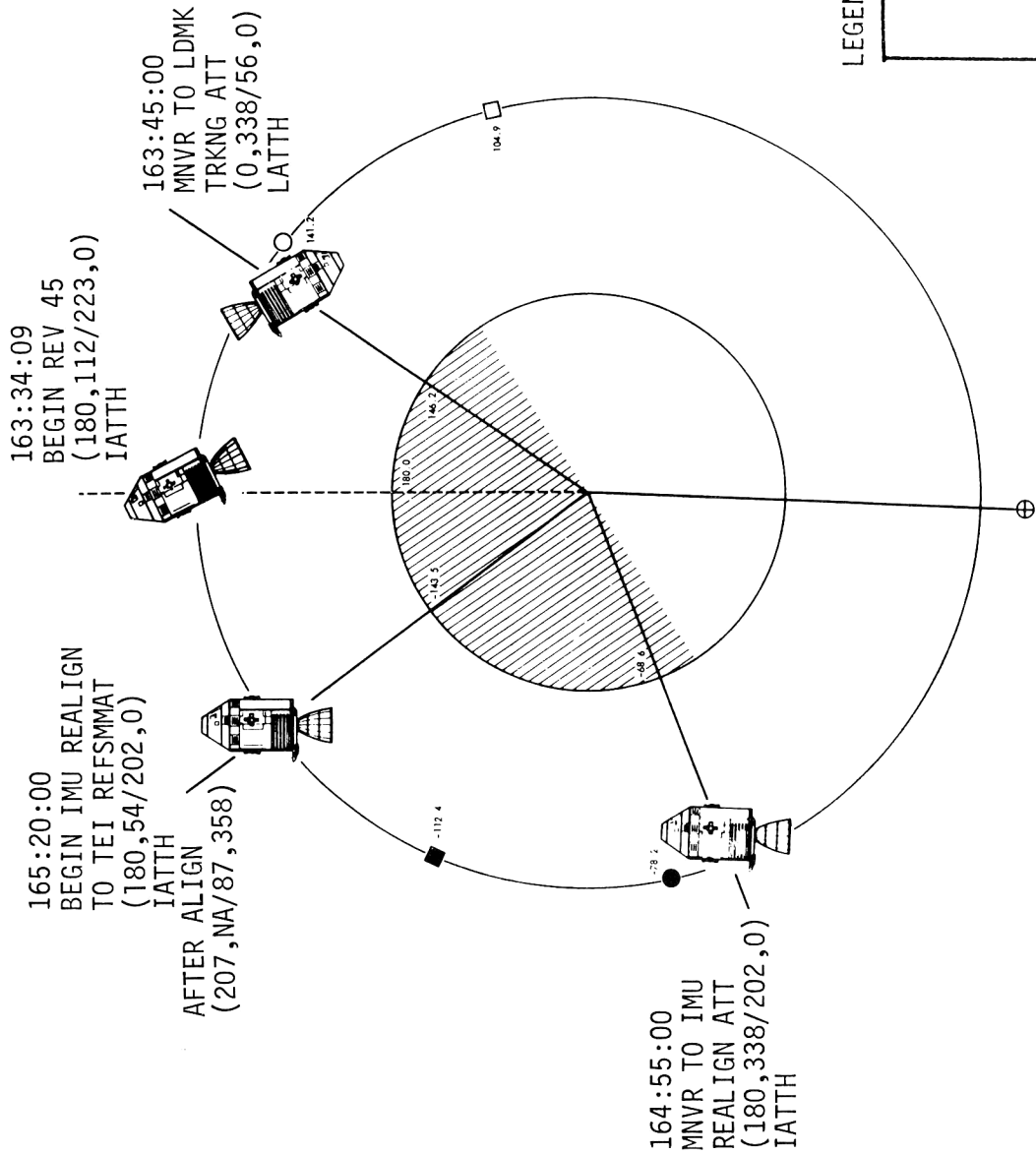
REV 43

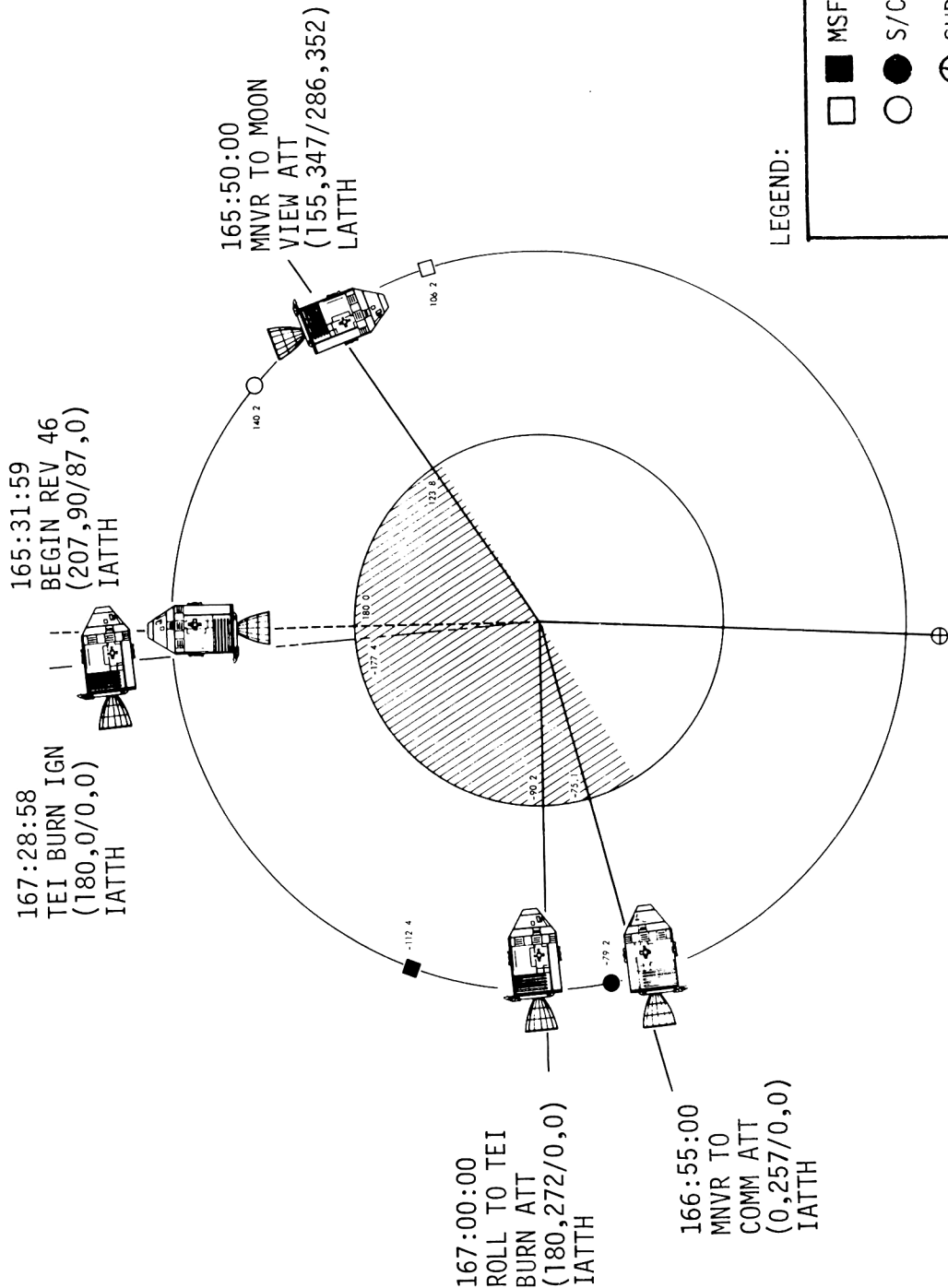




LEGEND:

REVISION A





LEGEND:

□	MSFN AOS, LOS
○	S/C SUNRISE, SUNSET
⊕	SUBEARTH POINT
(R,LHP/INP,Y).*	
IATTH - INERTIAL ATTITUDE HOLD	
LATTH - LOCAL ATTITUDE HOLD	

REVISION A

